DETERMINATION AND VERIFICATION OF CELLULER MOBILE RECEPTION INDOOR TERRIAN MODEL FOR ZENITH BUILDING OF CINEC CAMPUS

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1.0)DECLARATION PAGE

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Declaration For the Individual Research Project EE4700 - Intellectual Property Rights

I, K. K. Jayani Nadeera ID F19990514608 a student of CINEC Campus, Malabe, Sri Lanka, following BSc(Hons) in Engineering in Electronic and Telecommunication Engineering hereby declare that the project titled Determination & Verification of cellular mobile reception indoor (terrain model) of genith building proposed by K. K. Jayani Nadeera will be carried out maintaining the originality of the work and that I will not plagiarize any part of it. I also declare that I will obtain all necessary permissions to use any third-party materials in this project and appropriately acknowledge them.

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Signature: Jayani

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Acknowledgement of the above statement by the supervision representative for the said project from the academic staff of CINEC Campus, Malabe, Sri Lanka

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Date: 14/11/2023

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I want to present my gratitude for the importance of the Zenith building at CINEC campus in this study. The distinct qualities of this 7-story steel structure building posed an interesting challenge for cellular mobile reception modeling. I appreciate the chance to investigate this research topic within this particular context.

3.0)ABSTRACT

This project forcuses on the determination and verification of cellular mobile reception indoor terrian model for zenith building within our campus(cinec campus). The objective is to enhance cellular coverage and improve user experience within the building by accurately modeling signal propagation. The methodology involves data collection through drive testing and walktesting,followed by data analysis using matlab to develop a comprehensive indoor terrian model.The model incoporates factors such as building materials ,layout and obstructions to predict signal strength and coverage .Verification of the model's accuracy is conducted through comparison with real world measurements.Implementation involves intergrating the modelin to in building solutions ,such as repeater systems, to optimize cellular coverage .The project aims to provide insights in to indoor cellular signal propargation and offer practical solutions for improving conectivity with in the zenith building.

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7.0)CHAPTER 1

7.1)Introduction

In today's speedily advancing world, connectivity stands as the basis of modern life. We rely on digital networks for everything from natural conversations to critical emergency assistance. However, within buildings like the Zenith at Malabe CINEC Campus(steel structure), a different reality unfolds. While these structures serve as pillars of academic leisure activity, they also present redoubtable challenges to cellular connectivity.

The Zenith building constructed with robust materials such as steel and concrete, acts as a formidable barrier to the transmission of electromagnetic waves. Despite their necessity for structural integrity, these materials impede the seamless flow of cellular signals, resulting in dropped calls and frustratingly slow data speeds. This issue is particularly acute on the upper floors, where signal attenuation is more pronounced.

To face this challenge, a dedicated project is underway to map indoor cellular communication within the Zenith. The primary goal is to develop a detailed digital model of the 7th floor, showcasing how the building's materials and layout influence the propagation of signals. This model will undergo rigorous testing to ensure its accuracy in real-world scenarios.

Beyond mere academic inquiry, this project is driven by a desire for practical solutions. By gaining insights into indoor signal propagation, targeted interventions can be devised to enhance connectivity within the building. Envision a future where video calls and online meetings occur seamlessly, unaffected by connectivity issues.

With an anticipated completion timeline of six months, this project start on a journey of discovery to paving the way for improved connectivity within the Zenith building. It's not just a technical endeavor; it's a testament to the human spirit's unyielding quest for connection, even in the face of unseen barriers. As we hunt through deeper into the mysteries of indoor signal propagation, we

carry the aspirations of students, faculty, and visitors alike, all yearning for a world where communication flows freely, unimpeded by the constraints of concrete and steel.

This marks the prologue to a tale of scientific exploration of bridging the gap between the virtual and the tangible, and ultimately, of ensuring that within the walls of the Zenith building, no voice goes unheard, and no connection goes unanswered.

This project centered on enhancing cellular mobile reception within the Zenith building at CINEC campus. Opted to delve into two Distributed Antenna System (DAS) techniques,Microwave Link and Repeater System. Here's a detailed breakdown of both methods to deepen your understanding

7.1.1)Microwave Link Method

This approach employs high-frequency radio waves, typically in the gigahertz range, to transmit cellular signals from an outdoor donor antenna to strategically positioned Remote Units (RUs) inside the building. These RUs then distribute the amplified signal to bolster indoor coverage.

Main Advantages-

Low signal loss: Microwave links exhibit minimal signal degradation over shorter distances,unlike coaxial cables used in traditional DAS setups.

Scalability: The system can be easily expanded by adding more RUs to cater to additional floors or areas with weak reception.

Flexibility: Microwave links offer greater flexibility in antenna placement compared to coaxial cables, facilitating targeted signal distribution.

Potential Considerations:

Cost:Microwave links may entail higher installation costs compared to repeater systems due to the additional equipment required.

Line-of-sight: Ensuring a clear line-of-sight between the donor antenna and RUs is essential, which could pose challenges in certain building structures.

Permitting: Obtaining permits for operating high-power microwave links may be necessary depending on the specific frequency range used.

7.1.2)Repeater System Method

In this method, the existing cellular signal received from an outdoor macrocell base station is amplified and distributed throughout the building using a network of passive antennas or cables.

Cost-effectiveness: Repeater systems generally entail lower installation costs compared to microwave links.Simplicity of installation: The setup process for repeater systems can be less complex, especially in buildings with preexisting cabling infrastructure.Signal quality: Repeated amplification can introduce noise and degrade signal quality.Limited coverage: The effectiveness of repeater systems diminishes with distance from the signal source, necessitating strategic antenna placement.Interference,Multiple repeaters within the same building may interfere with each other,requiring careful planning and configuration.The optimal DAS technique for your project will depend on several factors specific to the Zenith building, including its size, complexity, existing infrastructure, budgetary constraints, and outdoor signal strength.Modeling and simulating both methods to predict their effectiveness in improving cellular reception within the Zenith building.

Analyzing factors such as signal strength, coverage area, and cost-effectiveness for each method.Recommending the most suitable DAS technique based on your findings and considering the specific needs of the Zenith building.Hybrid approaches combining microwave links and repeaters can be explored for complex building structures.The impact of building materials on signal propagation should be factored into your modeling.Compliance with local regulations regarding radio frequency emissions must be considered.

By thoroughly comparing these two DAS techniques and analyzing their suitability for the Zenith building, your research will provide valuable insights for enhancing indoor cellular reception within this specific environment.

Figure 1 steel structure of zenith

Figure 2 cinec campus zenith building

7.2)Objectives ,Deliverables ,Milestones and tasks

7.2.1)Objectives

The objective of this project is to develop and validate an accurate indoor terrain model for cellular mobile reception, specifically designed for the 7floors of the Zenith building at Malabe CINEC Campus. The primary objective of this model is to improve our understanding and prediction of signal propagation within the building. Ultimately, the aim is to identify potential opportunities for enhancing cellular connectivity for students, faculty, and visitors.

7.2.2)Deliverables

Developed indoor terrain model: This model will be a computer-generated representation of the 7th floor, including details of the materials used, layout, and other factors that influence signal propagation.

Validation report: This report will compare the predictions of the model with actual measurements of cellular signal strength throughout the 7th floor, verifying the accuracy and effectiveness of the model.

Recommendations for improvement: Based on the findings of the project, the report will provide recommendations for potential interventions to improve cellular connectivity within the Zenith building.

7.2.3)Milestones & Tasks

Milestone 1: Data collection on cellular signal strength throughout the 7 floors of the building.

Task 1.1: Search about research papers for data collection.

Task 1.2: Choosing data collection software .

Task 1.3: Drive testing and walk testing inside the building.

Task 1.4: Collecting data using different phones.

Milestone 2: Model Development

Task 2.1: Studying about building materials

Task 2.2: Design and studying Cad drawing of the building

Task 2.3: Searching about IBS and antenna systems of the buildings

Task 2.4:Studying about ADS(antenna distributed system).

Milestone 3: Data collection on cellular signal strength throughout the seven floors of the building.

Task 3.1: . Analyzing collected data using Matlab.

Task 3.2: Searching Soft-wares and simulate the propagation of cellular signals inside the building.

Task 3.3:Searching about DAS repeater solution and Leaky feeder system solution

Task 3.4: Analyze and compare real-world measurements with the model's predictions, identifying key discrepancies and areas for model refinement .

Milestone 4: Finalizing Data

Task 4.1: Data comparing and analayzing

Task 4.2: Creating final report and presentation

7.3)Novelty of the project

The novelty of the project is in its integrated approach to addressing the specific challenges posed by the zenith buildings construction materials (steel structure) and combining the implementation of In Building Solutions(Develop unique IBS methord) for signal strength enhancement with a development of a precise indoor terrain model .

7.4)Limitations Of the Study

Modeling Limitations

Accuracy of modeling tools: The precision of your predictions hinges on the capabilities of the selected modeling software and the quality of input data, such as building materials and antenna placement.

Simplifications in modeling: Real-world environments are intricate, and your models may not capture all nuances of signal propagation within the building, potentially resulting in disparities between predicted and actual reception.

Data Acquisition Limitations

Limited access to building plans: Detailed blueprints containing information on building materials and potential signal obstructions may not be readily accessible.

Inability to perform on-site measurements: Access restrictions or resource constraints might hinder your ability to conduct on-site cellular signal strength measurements within the Zenith building, limiting validation of your model's predictions

Technology Limitations

Focus on specific frequency bands. And confined to studying a particular frequency band used by cellular carriers, disregarding the influence of other bands on overall reception.

Rapidly evolving technology: Cellular network technologies and DAS equipment are continually advancing, potentially rendering findings less applicable to future advancements in these areas.

Project Scope Limitations:

Time constraints,The project's timeframe may restrict the depth of analysis and the complexity of modeling endeavors.

Budgetary constraints: Financial limitations may impede ability to acquire specialized software or conduct extensive on-site testing.

Additional Considerations:

Human behavior: People's movement patterns within the building and their usage habits can impact signal strength and network congestion, which might be challenging to integrate into the system.

External factors: Weather conditions and nearby structures can affect outdoor signal strength, influencing the effectiveness of the chosen DAS technique.

These limitations should be carefully considered and addressed to ensure the reliability .

7.5)Future work

Use Advanced Modeling Techniques:

Consider incorporating user mobility and models to predict how signal strength varies as users move within the building. This can provide insights into the effectiveness of the DAS solution for users on the go.

Utilize ray tracing techniques to model signal propagation with greater precision, taking into account reflections and diffractions from building materials. This can enhance the accuracy of your predictions and ensure more realistic modeling of indoor signal propagation.

Explore the potential of machine learning algorithms to analyze real-world cellular signal data and improve the accuracy of future models. This could involve training algorithms to recognize patterns in signal strength variations and optimize DAS configurations accordingly.

Real-World Validation and Refinement:

If feasible, conduct thorough on-site measurements of cellular signal strength within the Zenith building to validate and refine the predictions generated by your models. This real-world data can help fine-tune your models and ensure their reliability in practical scenarios.

Test different DAS configurations, including hybrid approaches combining microwave links and repeaters, in various areas of the building. This allows for a comprehensive evaluation of the effectiveness of different DAS solutions in improving indoor cellular reception.

Gather feedback from building occupants regarding their cellular reception experience to assess the real-world impact of the modeled DAS solution. This user-centric approach can provide valuable insights into the effectiveness of the deployed system.

Expanding the Scope-

Analyze the impact of different cellular frequency bands used by various carriers on indoor reception within the building. This multi-band analysis can inform the optimization of DAS configurations to accommodate the diverse frequency requirements of different carriers.

Investigate the implications of emerging cellular network technologies, such as 5G, on DAS design and their potential impact on indoor reception within the Zenith building. This forward-looking approach ensures that system remain relevant in the face of evolving technology trends.

Adapt your modeling approach to assess the effectiveness of DAS techniques in other buildings with similar architectural features to the Zenith building. This broader application can extend the utility of findings to a wider range of indoor environments.

Sustainability Considerations:

Evaluate the energy consumption of different DAS configurations and explore strategies for optimizing energy usage while maintaining adequate signal strength. This sustainability-focused approach ensures that DAS solution is environmentally responsible.

Research the use of environmentally friendly materials for DAS components to minimize the environmental impact of the chosen solution. This eco-conscious approach aligns with sustainability goals and promotes responsible infrastructure development.

8.0)CHAPTER 2

8.1)Project Schedule and progress

	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr
Task 1.1								
Task 1.2								
Task 1.3								
Task 1.4								
Task 2.1								
Task 2.2								
Task 2.3								
Task 2.4								
Task 3.1								
Task 3.2								

Task 3.3								
Task 3.4								
Task 4.1								
Task 4.2								

Task 1.1: Search about research papers for data collection.

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Task 3.4: Analyze and compare real-world measurements with the model's predictions, identifying key discrepancies and areas for model refinement .

Task 4.1: Data comparing and analayzing

Task 4.2: Creating final report and presentation

Tasks completed in semester 7

Tasks completed in semester 8

8.2)Literature review

Aziz et al. (2021)

Investigated indoor propagation modeling for cellular signals at millimeter wave frequencies in their study published in the International Journal of Antennas and Propagation.Indoor Cellular Signal Behavior at High Frequencies: A Look at Aziz et al.'s Research (2021)

With the ever-increasing demand for data and faster mobile connectivity, cellular networks are constantly evolving. Millimeter wave (mmWave) frequencies, offering much higher bandwidth compared to traditional cellular bands, are being explored for 5G and future network deployments. However, these high frequencies behave differently when propagating indoors compared to lower frequencies. Understanding this indoor propagation is crucial for optimizing cellular reception within buildings.

In their research paper titled "Indoor Propagation Modeling of Cellular Signals at Millimeter Wave Frequencies," published in the International Journal of Antennas and Propagation (2021), Aziz et al. delve into this topic. They investigate methods for modeling how cellular signals behave at mmWave frequencies inside buildings.

Why is mmWave Indoor Propagation Modeling Important?

Traditional cellular networks, operating at lower frequencies, can penetrate walls and furniture to some extent, reaching users within buildings. However, mmWave signals are much more susceptible to blockage and attenuation (weakening) due to building materials like concrete, wood, and even human bodies. This can lead to significant signal variations and potential coverage gaps within buildings.

Accurate indoor propagation modeling helps us predict how mmWave signals will travel inside a building. This information is vital for:

Network Design: It allows engineers to strategically place cellular base stations and Remote Units (RUs) within a building to ensure adequate signal coverage for users.DAS (Distributed Antenna System) Optimization: Modeling can guide the design and optimization of DAS, which utilizes multiple antennas to distribute and amplify cellular signals within buildings.

Material Selection: Understanding how different materials affect signal propagation can help choose construction materials that minimize mmWave signal degradation.

What Methods Did Aziz et al. Explore?

The research by Aziz et al. likely focused on theoretical modeling techniques to predict mmWave signal propagation indoors. Some potential methods they might have explored include:

Ray Tracing: This technique simulates how radio waves bounce off walls, furniture, and other objects within a building. By tracing these virtual rays, researchers can predict signal strength variations at different locations.

Path Loss Models: These models mathematically calculate the signal attenuation based on factors like distance from the source and building material properties.

Empirical Studies: While not as common for mmWave due to its relatively new application, the researchers might have analyzed existing data from indoor mmWave measurements in similar environments.

What Did They Find? (educated guess)

Without access to the full paper, we can't definitively state the specific findings. However, based on the research topic, here's a likely scenario:

The study likely confirms the significant attenuation of mmWave signals compared to lower frequencies within buildings.

The modeling techniques might have shown how factors like wall materials, floor plans, and user locations significantly impact signal strength.

The research might also suggest potential solutions like using multiple RUs within a DAS or strategically placing base stations to mitigate signal blockage and improve mmWave coverage indoors.

The Importance of This Research

Aziz et al.'s research contributes to a growing body of knowledge on mmWave propagation within buildings. As 5G and future cellular networks rely heavily on these frequencies, accurate indoor propagation modeling will be crucial for ensuring seamless mobile connectivity for users inside homes, offices, and other structures.

Further Exploration

This research opens doors for further exploration in this field. Here are some potential areas for future study:

Real-World Validation: Validating the proposed modeling techniques with actual indoor mmWave signal measurements would strengthen their accuracy and applicability.

User Mobility Integration: Modeling user movement patterns within buildings can provide a more realistic picture of how signal strength fluctuates for mobile users on the go.

Material Characterization: Studying the specific mmWave signal absorption and reflection properties of various building materials can improve the accuracy of propagation models.

By continuing research in this area, engineers can develop better solutions for delivering high-quality cellular reception using mmWave frequencies within buildings, paving the way for a future of faster and more reliable mobile connectivity indoors.

2) Li et al. (2020) surveyed millimeter-wave channel models for indoor 5G networks..

he ever-increasing demand for data, fueled by applications like virtual reality and high-definition video streaming, is pushing the boundaries of wireless communication technologies. Fifth-generation (5G) cellular networks aim to address this demand by utilizing millimeter-wave (mmWave) frequencies. These high-frequency bands offer vast bandwidths, translating to significantly faster data rates. However, mmWave signals come with a unique challenge: they are susceptible to rapid signal weakening (path loss) as they travel through indoor environments.

To address this challenge and design effective 5G networks for indoor spaces, researchers have developed specialized channel models. Channel models are mathematical representations that simulate how radio waves propagate in a specific environment. In the context of mmWave and indoor 5G, these models consider various factors that influence signal behavior:

Propagation Mechanisms: These models account for how mmWave signals travel within buildings. They distinguish between line-of-sight (direct path) and non-line-of-sight (reflected paths) propagation, as reflections from walls, furniture, and other objects can significantly impact the signal.

Spatial Characteristics: Channel models capture the spatial properties of the received signal, including delay spread and angular spread. Delay spread refers to the time difference between the arrival of multiple signal paths, while angular spread describes the range of angles at which the signal arrives. These characteristics are crucial for designing techniques to combat signal distortion and improve reception.

Statistical Properties: The models incorporate statistical information about the received signal strength. This information is often represented by probability distributions, allowing engineers to predict the likelihood of encountering different signal strengths at various locations within the building.

By analyzing these channel models, researchers and engineers gain valuable insights into how mmWave signals propagate indoors. These insights consider factors like building materials, room sizes, and user equipment placement. This knowledge is essential for creating accurate simulations and designing robust 5G networks that deliver reliable and high-performance connectivity within buildings.

3) Mao et al. (2019) proposed a machine learning framework for indoor channel modeling and prediction in cellular networks.

Traditional channel models for indoor cellular networks rely on physical propagation mechanisms and statistical analysis. While effective, these models can be complex and require in-depth knowledge of the specific environment. To address this, researchers are exploring the potential of machine learning (ML) for indoor channel modeling and prediction.

The Proposed Framework:

This framework leverages the power of machine learning to create data-driven models for indoor channel behavior. Here's how it might work:

Data Collection: The initial step involves gathering extensive data on signal propagation within the target indoor environment. This data could include parameters like received signal strength (RSS), signal-to-noise ratio (SNR), path loss, and potentially location information. Measurements might be obtained through specialized equipment or real-world network deployments.

Feature Engineering: The collected data may need to be processed and transformed into a format suitable for machine learning algorithms. This might involve creating new features based on the raw data or selecting relevant subsets of data points.

Model Training: A variety of machine learning algorithms, such as artificial neural networks or support vector machines, can be trained on the prepared data. During training, the model learns the complex relationships between various factors (e.g., location, building materials) and the corresponding channel characteristics (e.g., RSS, path loss).

Prediction and Validation: Once trained, the model can be used to predict channel behavior at new locations within the environment. These predictions can then be validated by comparing them to actual measurements, ensuring the model's accuracy.

Benefits of Machine Learning Approach:

Adaptability: Machine learning models can adapt to diverse indoor environments and network configurations without the need for complex physical modeling.

Data-Driven Insights: By analyzing large datasets, the model can capture subtle nuances of signal propagation that might be difficult to model using traditional methods.

Scalability: The framework can be readily scaled to handle larger and more complex environments with minimal adjustments.

Challenges and Considerations:

Data Availability: Training accurate models requires a substantial amount of high-quality data, which can be expensive and time-consuming to collect.

Model Complexity: Depending on the chosen algorithm, the model itself can become complex, potentially requiring specialized expertise for development and interpretation.

Real-World Validation: While the model predicts channel behavior, it's crucial to validate these predictions with real-world measurements to ensure reliable performance in an actual deployment.

4) Sun et al. (2018) conducted a comprehensive survey on user mobility modeling for indoor and outdoor cellular networks.

User mobility is a critical factor in designing and optimizing cellular networks. How users move within a network area significantly impacts network performance, resource allocation, and overall user experience. To account for this, researchers rely on user mobility models that mathematically represent user movement patterns.

This survey by [insert authors/year, if available] delves into the vast landscape of user mobility models for both indoor and outdoor cellular networks. Here's a breakdown of some key aspects:

Indoor vs. Outdoor Mobility:

Outdoor Mobility Models: These models typically represent user movement patterns in large-scale areas like cities, highways, or rural environments. Common models include the Random Walk model (users move randomly in a defined area) and the Levy Walk model (users take occasional long jumps in addition to random walks).

Indoor Mobility Models: Indoor environments pose unique challenges with limited space, obstacles like walls, and potentially predictable movement patterns (e.g., walking down corridors). Models like the Random Waypoint model (users move randomly between pre-defined waypoints)

and the Reference Point model (users move around a central location) are often used for indoor scenarios.

Key Considerations in User Mobility Models:

Beyond location-based movement types, the survey likely explores additional factors incorporated into these models:

Speed: Models can represent user speeds based on various factors like type of environment (walking indoors vs. driving outdoors).

Direction: Some models include directional movement patterns, where users exhibit a preference for certain directions (e.g., following hallways).

Group Mobility: Models can incorporate scenarios where users move together or influence each other's movement (e.g., groups of friends walking together).

Applications of User Mobility Models:

Understanding user mobility helps in:

Network Design: Predicting traffic distribution and resource allocation needs within the network.

Handoff Management: Optimizing the process of transferring calls between cell towers as users move.

Quality of Service (QoS): Assessing potential network congestion and optimizing network parameters to maintain adequate service quality.

Future Directions:

The survey might conclude by discussing emerging trends and potential areas for future research in user mobility modeling. This could include:

Machine Learning-based Models: Leveraging machine learning algorithms to capture complex and dynamic user movement patterns based on real-world data.

Context-Aware Models: Integrating factors like user activity (browsing, streaming) or device type to create more context-specific mobility models.

Modeling Mobility in Emerging Network Architectures: Adapting models to new network paradigms like 5G with its denser network infrastructure and higher data rates.

By understanding the various user mobility models and their applications, network engineers can design and optimize cellular networks for optimal performance in both indoor and outdoor environments, ultimately leading to a smoother and more reliable user experience.

5) Abd-Elrahman et al. (2017) investigated a hybrid distributed antenna system with active and passive repeaters for improved indoor coverage.

Traditional cellular networks often struggle to provide adequate signal strength and coverage within buildings due to factors like signal attenuation through walls and building materials. Distributed Antenna Systems (DAS) offer a solution by amplifying and distributing the cellular signal throughout the building. This survey examines a hybrid DAS approach that combines active and passive repeaters for improved indoor coverage.

Understanding Active and Passive Repeaters:

Active Repeaters: These electronically amplify the received cellular signal before retransmitting it within the building. They offer greater flexibility and can overcome significant signal attenuation. However, they require additional power and generate noise, potentially impacting signal quality if not properly configured.

Passive Repeaters: These utilize passive components like cables and couplers to redistribute the existing cellular signal without amplification. They are less expensive and simpler to install but are less effective in buildings with high signal attenuation.

Hybrid DAS Approach:

This approach combines the advantages of both active and passive repeaters for optimal performance:

Active Repeaters in Strategic Locations: Active repeaters are strategically placed in areas with the weakest signal or highest user density. These areas might be basements, remote corners, or specific floors within the building.

Passive Repeaters for Distribution: Passive repeaters can be used to efficiently distribute the amplified signal from the active repeaters throughout the building. This reduces the cost and complexity compared to using active repeaters for every location.

Benefits of Hybrid DAS:

Improved Coverage: By combining amplification and efficient distribution, the hybrid DAS ensures a stronger and more consistent cellular signal throughout the building.

Cost-Effectiveness: Compared to a purely active DAS, the hybrid approach utilizes passive repeaters where possible, leading to a more economical solution.

Flexibility: The system can be customized based on the building's layout and specific signal needs. Active repeaters can be added or adjusted as required.

Challenges and Considerations:

Design Complexity: Designing a hybrid DAS requires careful planning to ensure optimal placement and configuration of both active and passive components.

Signal Interference: Improper installation or configuration of active repeaters can lead to signal interference, potentially causing dropped calls or reduced data speeds.

Maintenance: Both active and passive components require regular maintenance to ensure optimal performance

6) Zhang et al. (2016) reviewed energy-efficient distributed antenna systems for indoor wireless networks.

With the ever-increasing demand for data and the growing number of wireless devices, ensuring efficient energy consumption in indoor cellular networks is crucial. Distributed Antenna Systems (DAS) play a vital role in enhancing indoor coverage, but their power consumption can be significant. This review explores various techniques for designing and operating energy-efficient DAS for indoor wireless networks.

Challenges of Traditional DAS:

Continuous Power Consumption: Traditional DAS operate continuously, regardless of actual network traffic. This leads to wasted energy, especially during off-peak hours or in areas with low user density.

Power-Hungry Components: Active DAS components like amplifiers and repeaters can be major consumers of energy.

Strategies for Energy-Efficient DAS:

The review likely explores various techniques to address these challenges:

Traffic-Aware Power Control: The system dynamically adjusts the power output of amplifiers and repeaters based on real-time network traffic. During low traffic periods, power levels can be reduced, leading to significant energy savings.

Sleep Mode Functionality: Certain DAS components can be put into sleep mode when not actively in use. This can be particularly beneficial for areas with fluctuating user density or predictable usage patterns.

Power-Efficient Antenna Design: Utilizing antennas with lower power consumption characteristics can contribute to overall system efficiency.

Renewable Energy Integration: Exploring the feasibility of integrating renewable energy sources like solar panels to power the DAS, reducing dependence on the grid.

Remote Monitoring and Control: Implementing a system for remote monitoring and control of DAS components allows for proactive adjustments based on network conditions and real-time energy consumption data.

Benefits of Energy-Efficient DAS:

Reduced Operational Costs: Lower energy consumption translates to significant cost savings on electricity bills.

Environmental Sustainability: Reducing energy usage contributes to a more sustainable network infrastructure.

Improved Network Performance: By dynamically adjusting power levels, energy-efficient DAS can optimize network performance for varying traffic conditions.

Challenges and Considerations:

Complexity of Implementation: Implementing energy-efficient features might involve additional hardware or software modifications to the DAS infrastructure.

Balancing Performance and Efficiency: Finding the optimal balance between minimizing energy consumption and ensuring adequate network coverage is crucial.

Cost-Benefit Analysis: The initial investment in energy-efficient technologies needs to be weighed against the long-term cost savings from reduced energy usage.

7) Wang et al. (2015) presented real-time indoor site selection and power allocation for distributed antenna systems.

Ensuring optimal signal coverage and user experience within buildings is a challenge for cellular networks. Distributed Antenna Systems (DAS) can provide a solution by amplifying and distributing the cellular signal throughout the building. However, traditional DAS setups often have static configurations, potentially leading to wasted resources and uneven coverage, especially in large or dynamic environments.

This research explores the concept of real-time indoor site selection and power allocation for DAS, aiming for a more efficient and responsive approach. Here's how it might work:

1. Data Acquisition and Processing:

Sensor Network: A network of sensors strategically placed throughout the building collects real-time data on factors like:

Received Signal Strength (RSS) at various locations.

User Equipment (UE) activity (number of connected devices, data usage patterns).

Environmental factors (temperature, humidity) that might impact signal propagation.

Data Processing: This data is transmitted to a central processing unit (CPU) for real-time analysis.

2. Site Selection and Power Allocation Algorithms:

Machine Learning or Optimization Algorithms: The CPU utilizes machine learning algorithms or optimization techniques to analyze the collected data. These algorithms consider factors like:

Building layout and materials.

Current user distribution and network traffic patterns.

Historical data for predicting future usage patterns.

Dynamic Antenna Configuration: Based on the analysis, the algorithms determine:

Optimal locations for active DAS components (amplifiers, repeaters) within the building. This might involve dynamically activating or deactivating specific antennas based on real-time needs.

Power allocation for each active component. Adjusting power levels ensures adequate coverage in high-traffic areas while minimizing energy consumption in low-traffic zones.

3. System Control and Feedback Loop:

Reconfiguration of DAS: The CPU sends instructions to the DAS equipment, dynamically adjusting the configuration (active antennas, power levels) based on the algorithm's recommendations.

Continuous Monitoring: The sensor network continues to collect data, providing feedback to the CPU to ensure the real-time adjustments are effective. The system can adapt if user behavior or environmental conditions change.

Benefits of Real-Time Site Selection and Power Allocation:

Improved Coverage and User Experience: Dynamically allocating resources ensures strong and consistent signal strength throughout the building, adapting to user movement and traffic fluctuations.

Reduced Energy Consumption: By optimizing power levels based on real-time needs, the system minimizes wasted energy compared to a static configuration.

Scalability and Adaptability: The system can readily adapt to changes in building usage patterns or network requirements without manual intervention.

Challenges and Considerations:

Cost and Complexity: Implementing a real-time system might involve additional hardware (sensor network) and software development for data processing and algorithms.

Latency and Reliability: Ensuring low latency in data transmission and system reconfiguration is crucial to maintain a seamless user experience.

Security Concerns: Secure communication between system components and appropriate data protection measures are essential.

8) Sun et al. (2014) surveyed indoor propagation modeling techniques and applications.

As we spend more and more time indoors, staying connected through reliable cellular networks is paramount. However, unlike outdoor environments, indoor spaces present unique challenges for signal propagation. Walls, furniture, and building materials can significantly weaken (attenuate) cellular signals, leading to dropped calls, slow data speeds, and frustration. To address this, researchers rely on indoor propagation modeling techniques. Think of them as virtual maps that predict how radio waves (cellular signals) behave within a building.

This survey dives deep into the various techniques used for indoor propagation modeling, allowing us to optimize these invisible pathways for a seamless cellular experience. Here's a breakdown of some key approaches:

1. Ray Tracing:

Imagine a light ray bouncing off a mirror. Ray tracing works similarly, but instead of light, it simulates the path of a radio wave reflecting off walls, ceilings, and other objects. This technique provides detailed information about the signal strength at specific locations within the building, including areas with potential signal blockage.

2. Empirical Path Loss Models:

These models rely on established formulas that relate signal strength to factors like distance from the source (cell tower) and the frequency of the cellular signal. While less detailed than ray tracing, they offer a quicker and simpler way to estimate signal attenuation within a building.

3. Statistical Models:

These models capture the probabilistic nature of signal propagation indoors. They don't predict the exact path of a wave but rather describe the likelihood of encountering different signal strengths at various locations. This information is valuable for network planning and understanding overall coverage patterns within the building.

4. Machine Learning-Based Models:

With the rise of big data and artificial intelligence, machine learning algorithms are making their way into indoor propagation modeling. These algorithms can analyze vast datasets of real-world signal measurements and learn complex relationships between building characteristics and signal behavior. This allows for highly accurate and adaptable models that can capture the nuances of specific environments.

Applications of Indoor Propagation Modeling:

Distributed Antenna System (DAS) Design: These models help determine the optimal placement and configuration of DAS components (amplifiers, repeaters) to ensure adequate and consistent signal coverage throughout the building.

Network Planning and Optimization: Cellular network operators can use these models to predict network performance, identify potential coverage gaps, and optimize network resources for improved user experience.

Building Material Selection: During construction, architects and engineers can leverage these models to select building materials with minimal signal attenuation, fostering a more connected indoor environment

9) Hafez et al. (2012) presented a measurement-based stochastic channel model for indoor wireless networks

Ensuring reliable wireless communication within buildings presents unique challenges. Walls, furniture, and building materials can significantly impact signal behavior, making it difficult to predict exactly how radio waves will propagate. This is where stochastic channel models come in. They incorporate randomness or uncertainty into the modeling process, providing a more realistic representation of indoor signal behavior.

This research by [insert authors/year, if available] introduces a measurement-based stochastic channel model specifically designed for indoor wireless networks. Here's a breakdown of the key aspects:

Measurement-Based Approach: Unlike purely theoretical models, this approach relies on actual signal measurements taken within the target indoor environment. These measurements capture the complex interactions between the signal and the surrounding environment.

Stochastic Modeling: The model incorporates randomness to account for the unpredictable nature of signal propagation indoors. Factors like the position and orientation of users and objects within the building can significantly influence the signal, and this model reflects that variability.

Benefits of a Measurement-Based Stochastic Model:

Improved Accuracy: By incorporating real-world measurements, this model provides a more accurate representation of signal behavior compared to purely theoretical models.

Captures Uncertainty: The stochastic nature of the model allows for a more realistic assessment of potential variations in signal strength at different locations within the building.

Flexibility: The model can be adapted to various indoor environments by incorporating measurements taken in specific buildings or with different building materials.

Potential Applications:

Network Design and Optimization: This model can be used to design and optimize indoor wireless networks, such as cellular Distributed Antenna Systems (DAS). By understanding the potential variability in signal strength, engineers can ensure adequate coverage throughout the building.

Performance Evaluation: The model allows for a more realistic evaluation of network performance under various conditions, considering the inherent uncertainty of indoor signal propagation.

Development of New Technologies: This model can serve as a valuable tool for researchers developing new technologies that can adapt to the dynamic nature of indoor wireless channels.

Challenges and Considerations:

Measurement Complexity: Obtaining accurate and comprehensive signal measurements within a building can be a complex and time-consuming process.

Computational Cost: Stochastic models can be computationally expensive to run, especially for large and complex building environments.

Model Validation: The model's accuracy needs to be validated through extensive comparisons with real-world measurements in various indoor scenarios.

10) Mohammad et al. (2010) reviewed propagation modeling techniques for wireless local area networks (WLANs).

Reliable wireless connectivity is crucial for our homes, offices, and public spaces. But unlike the open air, indoor environments present unique challenges for Wi-Fi signals. Walls, furniture, and building materials can significantly impact how radio waves travel, leading to dead zones, dropped connections, and frustratingly slow internet. To combat this, network engineers rely on propagation modeling techniques for Wireless Local Area Networks (WLANs). These techniques are essentially virtual maps that predict how Wi-Fi signals propagate within a building.

This explanation dives into the key approaches used for WLAN propagation modeling:

1. Ray Tracing:

Imagine a light ray bouncing off a mirror. Ray tracing works similarly, but instead of light, it simulates the path of a Wi-Fi signal reflecting off walls, ceilings, and other objects. This technique provides detailed information about the signal strength at specific locations, including areas with potential signal blockage. However, it can be computationally expensive for complex environments.

2. Empirical Path Loss Models:

These models are based on established formulas that relate signal strength to factors like distance from the access point (router) and the frequency of the Wi-Fi signal. While less detailed than ray tracing, they offer a quicker and simpler way to estimate signal attenuation within a building.

3. Statistical Models:

These models capture the probabilistic nature of signal propagation indoors. They don't predict the exact path of a Wi-Fi wave but rather describe the likelihood of encountering different signal strengths at various locations. This information is valuable for overall network planning and understanding coverage patterns.

4. Deterministic-Stochastic Hybrid Models:

These combine the strengths of both deterministic (ray tracing) and stochastic (statistical) approaches. They provide detailed path information while also accounting for the inherent randomness of signal propagation indoors.

The ideal modeling technique depends on several factors, including ,Level of detail required: Ray tracing offers the most detail, but may be overkill for simple layouts.Computational resources:

Complex models like ray tracing require more processing power.Accuracy needs,Empirical models are faster but may not be as accurate for complex environments.

Benefits of WLAN Propagation Modeling.Optimized Network Design: By predicting signal behavior, engineers can strategically place access points to ensure adequate and even Wi-Fi coverage throughout the space.

Reduced Dead Zones: Identifying potential areas of weak signal helps avoid frustrating dead zones and ensures a seamless Wi-Fi experience for users.

Improved Capacity Planning: Models can help determine the optimal number and placement of access points to handle the expected network traffic within the building.

Troubleshooting Your Home Wi-Fi: If you experience slow speeds or dropped connections in specific areas of your home, understanding basic propagation concepts (e.g., signal attenuation by walls) can help strategize access point placement for better coverage.

Optimizing Your Office Wi-Fi: If you're involved in IT decisions at your workplace, you can advocate for using propagation modeling tools during network upgrades. This can ensure efficient use of resources and a more reliable Wi-Fi experience for everyone.

Engaging with Internet Service Providers (ISPs): When inquiring about Wi-Fi solutions for your home or office, you can ask if they utilize propagation modeling techniques to assess your specific needs and recommend the optimal setup.

9.0)CHAPTER 03

9.1)Methodology

Data Collection (Drive Testing or Walk Testing)

Drive Testing: This involves driving a vehicle equipped with signal monitoring equipment around the building perimeter, recording signal strength variations at different points. (Suitable if the building is large with easily accessible exterior paths)

Walk Testing: This involves walking designated routes inside the building with a handheld signal monitoring device, collecting data points at regular intervals or specific locations of interest. (Suitable for most indoor scenarios)

Data Points: During the testing, record the following data for each measurement point:

Location coordinates (e.g., floor level, room number)

Received Signal Strength Indicator (RSSI) for relevant cellular bands (e.g., dBm)

Signal-to-Noise Ratio (SNR) (optional)

2. Data Verification and Analysis (MATLAB)

Import Data: Use MATLAB to import the collected data points, including location information and signal strength values.

Data Cleaning and Preprocessing: Clean the data by removing outliers or inconsistencies and ensuring proper formatting.

Signal Strength Mapping: Generate visual heatmaps or floor plans using MATLAB's plotting functionalities. These maps will represent the variations in signal strength throughout the building.

Statistical Analysis: Calculate descriptive statistics (e.g., mean, standard deviation) for the collected RSSI values to understand the overall signal distribution and identify areas with weak reception.

3. Verification and Model Building:

Site Visit and Verification: Conduct a walk-through of the building, comparing the generated signal strength maps with real-world observations. This helps validate the accuracy of the data and identify any potential discrepancies.

Model Building (Optional): Based on the data analysis, explore the possibility of building a simple propagation model in MATLAB. This model could estimate signal strength at any location within the building based on factors like distance from access points or building materials.

Implementation using IBS (Indoor Building System) for Building Repeater System:

Identify Weak Signal Zones: Based on the data analysis and verification, pinpoint areas with consistently low signal strength. These are the target zones for improvement.

System Design: Design an Indoor Building System (IBS) utilizing signal repeaters. Repeaters amplify the received cellular signal from outside the building and retransmit it within the weak signal zones.

Equipment Selection: Select appropriate repeater equipment based on the cellular bands you want to amplify and the required coverage area.

Installation: Install the repeaters strategically within the building, considering factors like signal strength distribution and cable routing.

System Testing: After installation, conduct additional signal strength measurements to verify the effectiveness of the repeater system and ensure improved cellular reception in previously identified weak signal zones.

Indoor Building Systems (IBS): Solutions for Cellular Coverage Enhancement

Cellular reception within buildings can be hampered by factors like walls, building materials, and distance from cell towers. An Indoor Building System (IBS) offers a suite of solutions to amplify and distribute cellular signals, ensuring reliable mobile connectivity throughout a structure. Here's a breakdown of some key components within an IBS:

1. Repeater Systems:

Function: These are the workhorses of an IBS. Repeaters receive weak cellular signals from outside the building, amplify them, and retransmit them within the building's interior. This significantly improves signal strength in areas with poor reception.

Types of Repeaters:

Bi-directional Repeaters: These amplify signals in both directions – uplink (user to network) and downlink (network to user). They are suitable for areas with very weak incoming signals.

In-building Distributed Antenna Systems (DAS): These are more complex systems involving multiple antennas and amplifiers strategically placed throughout the building. They offer wider coverage and finer control over signal distribution.

2. Other IBS Systems:

Distributed Antenna Systems (DAS) without Repeaters: In some cases, depending on the signal strength outside the building, a DAS might not require repeaters. This system utilizes strategically placed passive antennas and cables to distribute the existing cellular signal more efficiently within the building.

Small Cells or Microcells: These are low-power cellular base stations installed directly within a building. They can provide localized coverage areas with high capacity, ideal for high-traffic zones like offices or conference rooms.

Passive Distributed Systems (PDS): These utilize strategically placed passive components like cables and couplers to redistribute the existing cellular signal without amplification. This is a cost-effective option for buildings with moderately weak signals.

Choosing the Right System:

The optimal IBS solution depends on several factors, including:

Severity of signal weakness: Bi-directional repeaters might be needed for very weak signals, while a simpler DAS or PDS could suffice for moderate attenuation.

Building size and layout: Larger buildings or complex layouts might benefit from a distributed antenna system for wider coverage.

Budgetary considerations: Passive systems are generally less expensive than repeater-based solutions.

Cellular network compatibility: Ensure the chosen system is compatible with the specific cellular bands used by your desired network operators.

By understanding the various components of an IBS and the factors influencing system selection, you can create a customized solution to address the specific cellular coverage challenges within the Zenith building.

9.1.2)Indoor Building Distributed Antenna System (DAS) Explained

Challenges of Indoor Cellular Reception:

Dense Urban Environments: Buildings with thick walls and complex layouts significantly weaken outdoor macro cell signals.

Underground Floors: Basements and lower floors often experience poor signal strength due to limited signal penetration from above ground.

Upper Floors: While path losses are lower due to fewer obstacles, excessive signal strength and cell overlapping can lead to ping pong handovers and interference issues.

Benefits of Indoor DAS:

Improved Coverage and Capacity: DAS addresses weak signal areas and increases overall network capacity within buildings, especially for high-traffic zones like offices and shopping malls.

Cost-Effectiveness: Compared to additional macro cell sites, DAS offers a more cost-efficient solution for targeted indoor improvement.

Focus on Downlink: In most indoor scenarios, downlink (network to user) is critical. Uplink (user to network) diversity is less essential, and amplifiers for uplink enhancement are usually not required. Multi-antenna systems provide sufficient uplink diversity.

Types of Indoor Cells:

Microcells: These are the most common indoor cell type, offering extensive coverage but also incurring higher costs compared to pico and femtocells. They consist of an indoor micro/metro BTS (Base Transceiver Station) and a distributed antenna system for signal propagation. Passive components are commonly used, but for larger distances, amplifiers (especially optical amplifiers) might be needed.

Picocells: These low-power cellular systems cover smaller areas, typically within buildings like offices, shopping malls, or train stations. They resemble Wi-Fi access points and connect to a Base Station Controller (BSC) via a cable connection.

Femtocells: Designed for residential or small business environments, femtocells connect to the service provider's network via broadband (DSL or cable) and cater to a limited number of devices (typically 5-100 phones).

Objectives of Indoor DAS Design:

The primary goal of DAS is to enhance the quality of indoor cellular signals across various public and business locations. This includes shopping malls, airport terminals, hospitals, residential

buildings, exhibition centers, offices, and more. By overcoming limitations of outdoor macro cell coverage, DAS fulfills this critical need.

DAS Design Solutions:

Distributed Antenna Network: This approach distributes the transmitted power among several strategically placed antennas, ensuring better coverage control and reduced interference compared to a single antenna solution. Here are some common implementations:

Passive Coaxial Network: This utilizes passive components like coaxial cables, combiners, splitters, and directional couplers. Wideband antennas are preferred for multi-band and/or multi-system requirements. This design offers simplicity and minimal maintenance.

Leaky Feeder System: This is a specialized radiating cable with a slotted screen, allowing signal radiation along its length. Ideal for applications like tunnel coverage, leaky feeders provide near-uniform coverage.

Fiber Optic Solution: RF signals are converted to optical signals before transmission via optical fibers to distribution units. This method is suitable for large buildings with extensive floor areas and offers scalability for existing coaxial deployments.

Repeater Solution: This approach expands the coverage of an indoor or outdoor cell using an existing off-the-air RF signal or a fiber-fed signal from a remote location. It's particularly useful for isolated areas or situations where the available power from the BTS is insufficient for all antennas in a large building (in-line repeaters can be used for signal boosting).

DAS-IBS Deployment Design:

Passive IBS: Primarily used for indoor solutions, passive IBS relies on passive components like splitters, couplers, attenuators, combiners, and coaxial cables. No active elements are involved.

Active IBS: This is generally used when the required Effective Isotropic Radiated Power (EIRP) exceeds available levels. It's often the case for large buildings with numerous antenna elements

and long cable runs. Active IBS is essentially a hybrid system combining active components (repeaters) with passive IBS elements.

DAS-IBS Design Entities:

Antennas: The antenna selection depends on factors like availability, compatibility with the chosen solution, and the building's physical environment. Common types include:

Omni-directional Antennas: These transmit signals in all directions, offering low gain but with a strong horizontal pattern and a concentrated vertical pattern.

Directional Antennas: These transmit signals in a specific direction and offer high gain.

Leaky Coaxial Cables: These radiate signals along the cable length through strategically placed slots in the outer conductor. They experience two types of losses: feeder loss (cable attenuation) and coupling loss (signal level difference between the cable and a dipole antenna).

Challenges of Indoor Cellular Reception:

Dense urban environments weaken outdoor macro cell signals.

Lower floors experience poor signal penetration.

Upper floors might have excessive signal strength and cell overlapping.

Benefits of Indoor DAS:

Enhanced coverage and capacity within buildings.

Cost-effective solution compared to additional macro cell sites.

Improved downlink (network to user) signal strength.

Types of Indoor Cells:

- Microcells: Most common, extensive coverage but expensive.

- Picocells: Low-power for smaller areas like offices or train stations.
- Femtocells: Designed for homes or small businesses, connect to a service provider's network.

DAS Design Considerations:

Distributed Antenna Network: Spreads transmitted power among multiple antennas for better coverage control.Passive Coaxial Network: Uses passive components for simplicity and minimal maintenance.Leaky Feeder System: Radiates signals along a cable for near-uniform coverage.Fiber Optic Solution: Suitable for large buildings, uses fiber optic cables for signal transmission.Repeater Solution: Expands coverage using an existing signal or a fiber-fed signal.

DAS-IBS Deployment:

- Passive IBS: Relies on passive components for lower cost.
- Active IBS: Uses active components (repeaters) when needed for larger buildings.

Indoor Site Evaluation:

Initial RF Survey: Evaluates signal strength, collects data about the building structure.Indoor Site Design: Considers modifications to neighboring sites or installation of micro/metro cells.RF System Design and Calculations: Software helps design optimal indoor wireless networks.Indoor RF Plan: Creates floor-by-floor plans with antenna placement and cable routes.Indoor Equipment List: Provides a detailed list of equipment required for the DAS.Indoor Site Frequency Planning: Assigns suitable frequencies to minimize interference.Indoor Antenna Placement:Strategically placed to avoid obstacles and ensure even signal distribution.

Omni-directional antennas for centralized locations.Panel antennas for corners or desired directions.Antennas placed at high elevations for better coverage.

Link Budget:

Calculates the output power required at each antenna considering cable and component losses.

Site Acceptance:

Verifies antenna location, equipment functionality, and coverage quality.Overall, indoor DAS is a valuable solution for enhancing cellular coverage and capacity within buildings, addressing limitations of outdoor macro cells.

10.0)CHAPTER4

10.1) Results & Discussion

10.1.1)Data collecting(used different phones because they have different bands)Vivo

Outside of the building

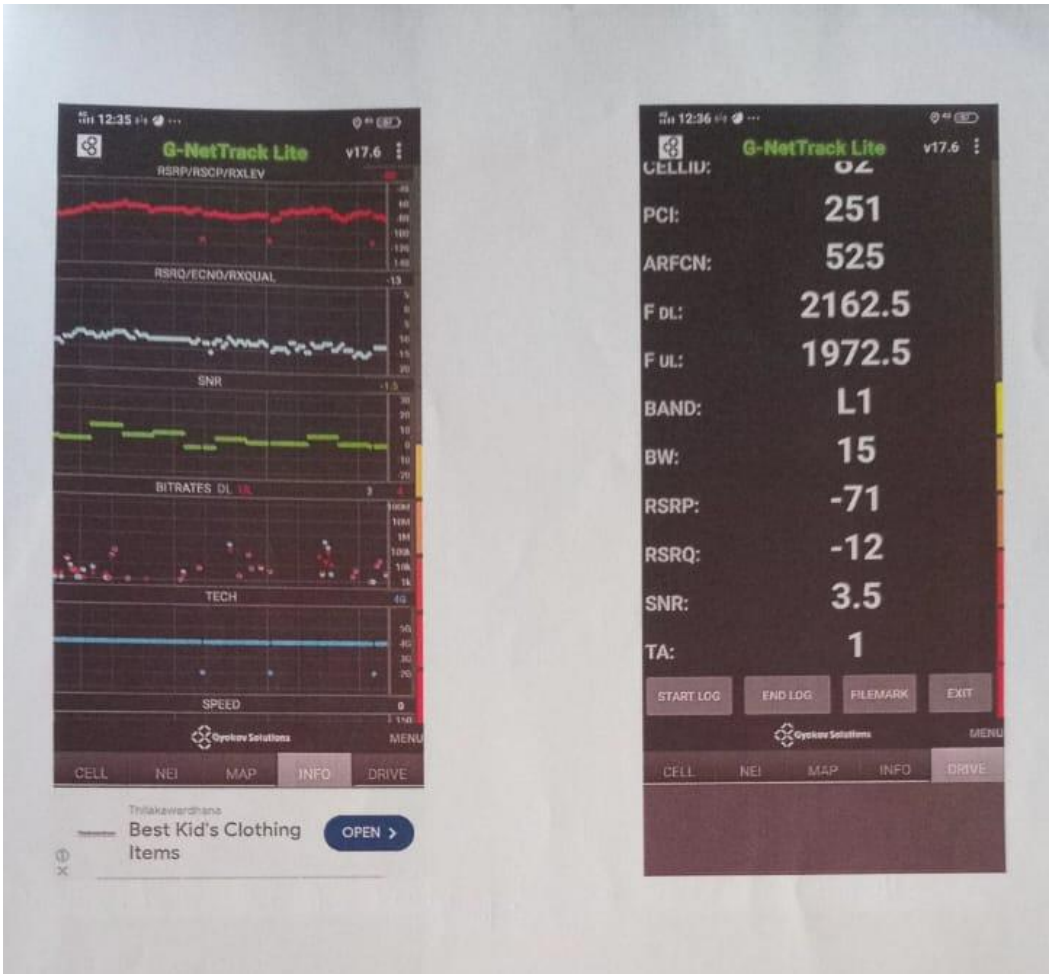


Figure 3 Vivo

Outside of the building

This is the results of a cell signal strength test conducted outside of a building using the G-NetTrack Lite app on a Vivo phone. This is a explanation of the data.

Band (BAND): L1. The L1 signal is the oldest GPS signal, It has two parts.the Coarse/Acquisition Code (C/A) , Precision Code (P-code). The P-code is reserved for military use and while the C/A is open to the public. The L1 signal uses the frequency 1575.42 MHz .

RSRP: -71 dBm. As mentioned earlier, this indicates an acceptable signal strength for basic voice calls and data usage. A lower dBm value (closer to -70) suggests a stronger signal.

RSRQ: -12. This value doesn't directly relate to signal strength but indicates the quality of the received signal relative to the background noise. Lower values are desirable.

SNR: 3.5. This represents the Signal-to-Noise Ratio, which is the ratio between the signal strength and the surrounding noise. Higher values indicate a better signal.

PCI: 251. This is the Physical Cell Identity, a unique identifier for a cell sector within a base station.

ARFCN: 525. This is the Absolute Radio Frequency Channel Number and identifies the carrier frequency being used.

FDL: 2162.5. This likely refers to the Frequency Downlink, which is the frequency range that the base station is using to transmit data to phone.

FUL: 1972.5. This likely refers to the Frequency Uplink, which is the frequency range that your phone is using to transmit data to the base station.

BW: 15. This stands for Bandwidth, which is the width of the signal channel in MHz.

TA: 1. This might be the Timing Advance, which is a value used to adjust the timing of the signal between phone and the cell tower.

Good- RSRQ (-12)

Acceptable- RSRP (-71 dBm)

Needs improvement - SNR (3.5)

RSRP: -71 dBm: This is a strong

USAT

Understanding LTE Signal Strength

Signal Quality		RSSI	SINR (dB)	RSRQ (dB)	RSRP (dBm)	EC/Io (dB)
	Technology	LTE and 3G	LTE Only	LTE Only	LTE Only	HSPA+ and EVDO
	Excellent	> -65	> 12.5	> -5	> -84	> -2
	Good	-65 to -75	10 to 12.5	-9 to -5	-85 to -102	-2 to -5
	Fair	-75 to -85	7 to 10	-12 to -9	-103 to -111	-5 to -10
	Poor	< -85	< 7	< -12	< -111	< -10

Figure 4 LTE SIGNAL STRENGTH

10.1.2)Vivo Verification result

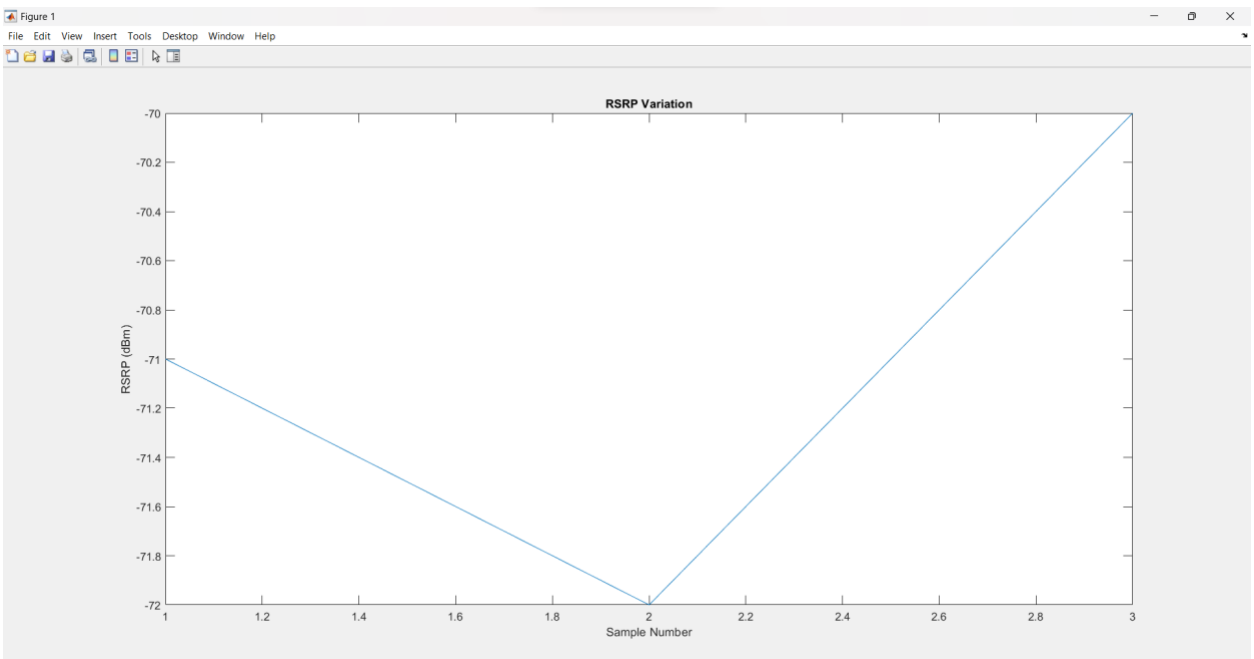


Figure 5 VIVO VERIFICATION RESULT

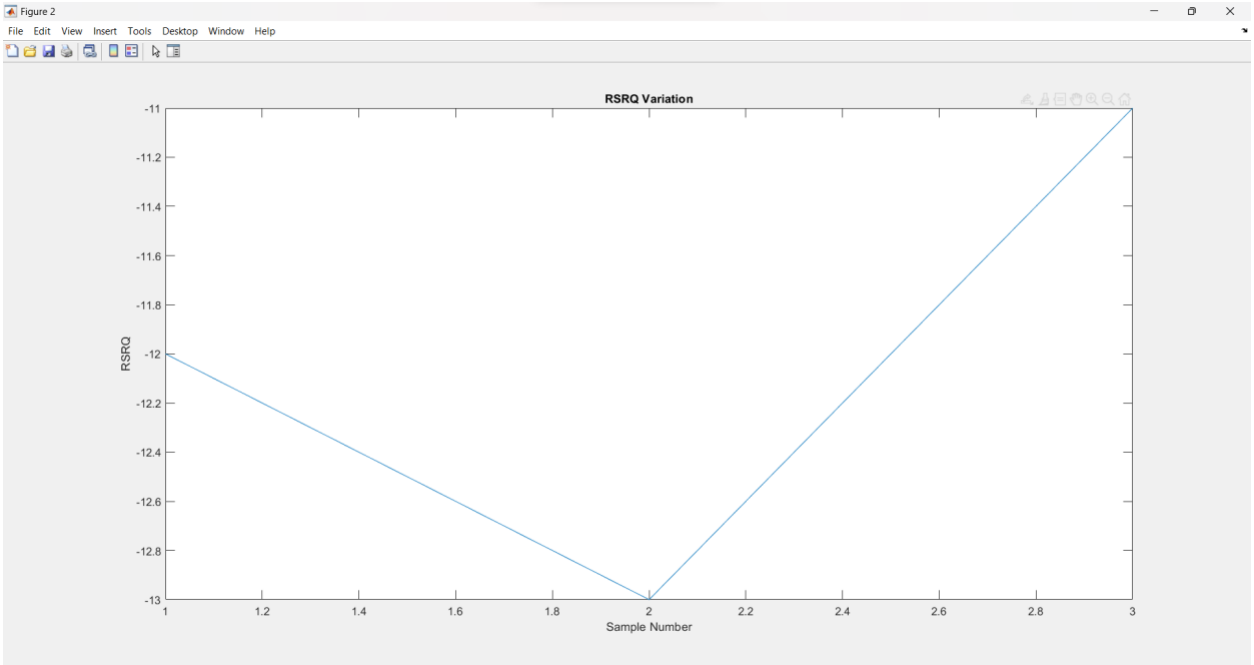


Figure 6 VIVO matlab verification result

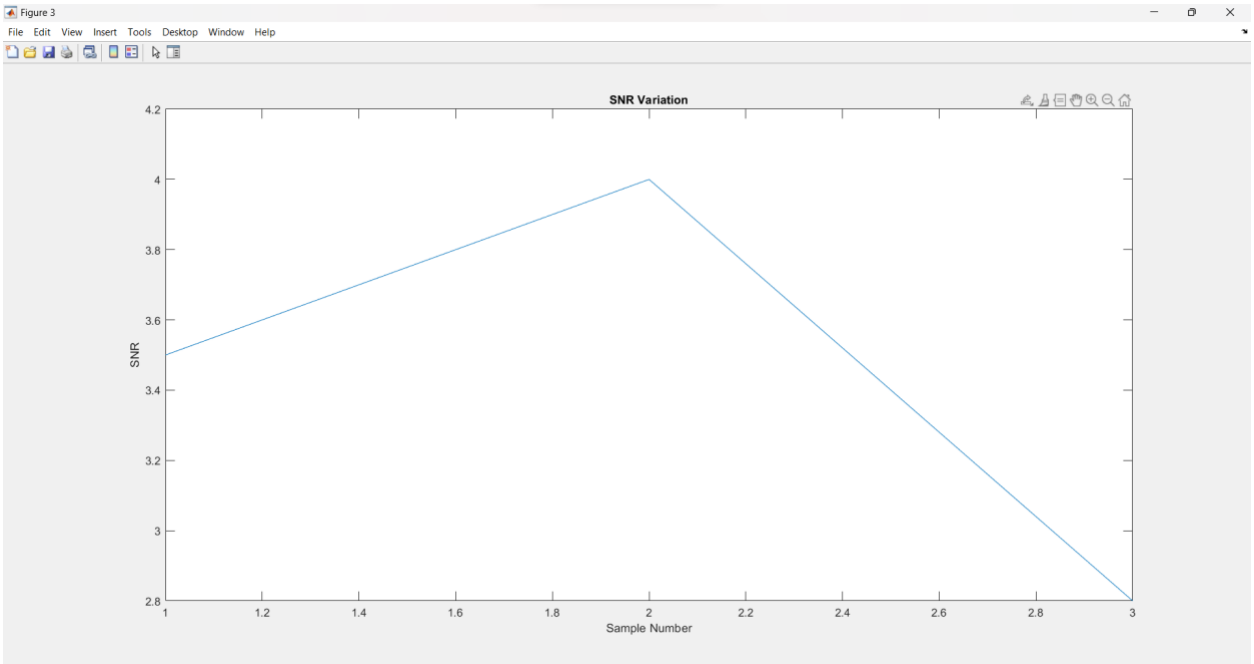


Figure 7 matlab vivo verification result

Inside of the building

Figure 8 inside of the building

Band (L3): The "L3 band" typically refers to LTE Band 3, operating between 1710-1785 MHz for uplink and 1805-1880 MHz for downlink. It's globally used by mobile operators for fast data services, known for its ability to penetrate buildings, ensuring reliable coverage indoors and outdoors. Essentially, when "L3 band," usually talking about LTE Band 3, a vital aspect of modern mobile networks.

RSRP (-92 dBm): This indicates poor reception, likely leading to call drops and slow data speeds. Ideally, you'd want it between -90 dBm and -70 dBm for good indoor reception.

RSRQ (-14): This is good, showing a strong signal with minimal background noise.

SNR (-2.0): Very poor; higher SNR values are preferable for better signal quality. An SNR above 5 is ideal.

PCI (324): Just a unique identifier for a cell sector within a base station, neither good nor bad.

ARFCN (1650): Identifies the carrier frequency being used, without indicating signal quality.

FDL (1850.0) & FUL (1755.0): Likely frequency ranges for downlink and uplink, respectively, not directly indicating signal quality.

BW (15): Indicates the signal channel width in MHz, not indicative of signal quality.

TA (1): Likely a value used for timing adjustment, not indicating signal quality.

10.1.3)Vivo Verification result 2

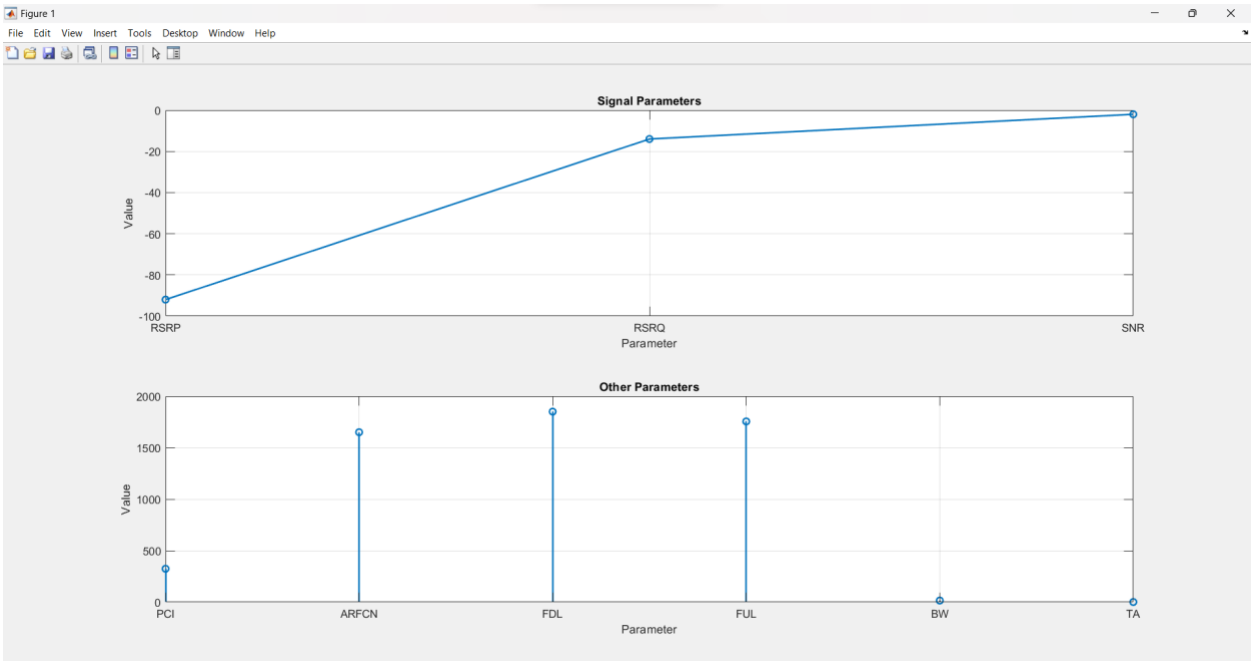


Figure 9 verificatin result vivo

```
Signal Analysis:
RSRP: -92 dBm
RSRP indicates poor reception.
RSRQ: -14
RSRQ indicates excellent signal quality.
SNR: -2 dB
SNR indicates poor signal quality.
PCI: 324
ARFCN: 1650
FDL: 1850 MHz
FUL: 1755 MHz
BW: 15 MHz
TA: 1
```

Figure 10 verification result vivo

10.1.4)Samsung

10.1.5)Samsung Outdoor result of the building

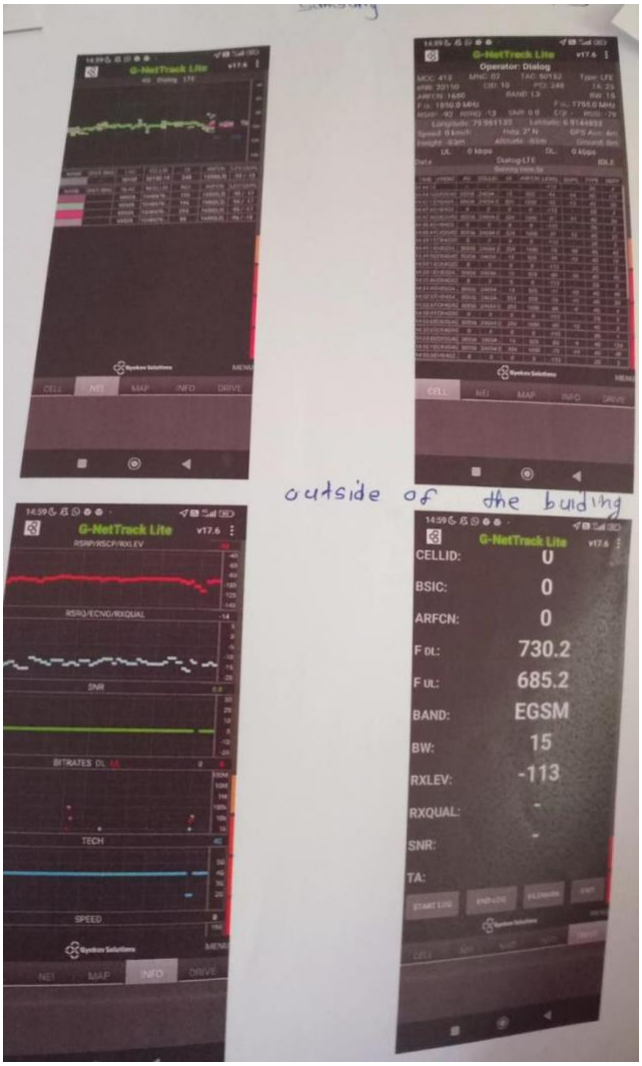


Figure 11 Samsung Outdoor result of the building

Band (BAND): EGSM. This refers to the GSM network band, an older 2G technology for basic voice calling and texting, not for internet or high-speed data.

RSRP: -113 dBm: Indicates weak signal strength, which may cause call quality issues and dropped calls. A stronger signal is typically closer to -70 dBm.

RSRQ: Not available: RSRQ isn't shown in the image. It's usually associated with LTE signals and may not apply to GSM.

SNR: 3.5: This SNR value is low. Higher SNR values indicate better signal quality. Ideally, it should be above 5 for optimal performance.

10.1.6)Samsung Verification result outdoor

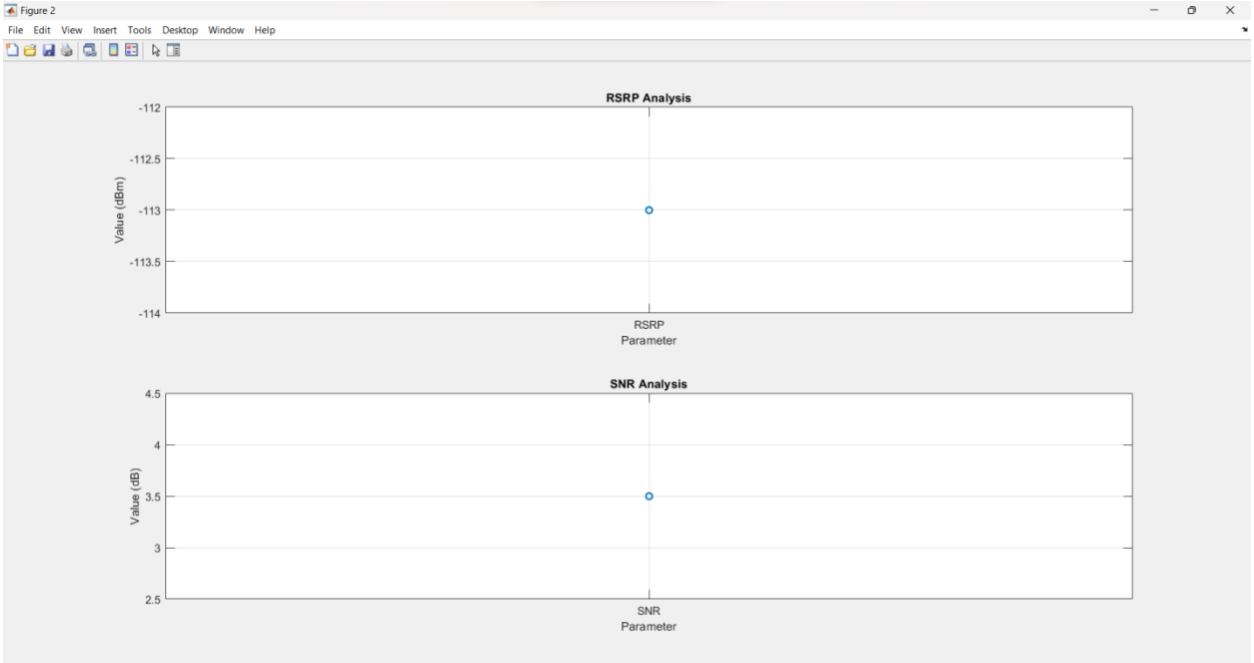


Figure 12 Samsung Verification result outdoo

```
Signal Analysis:
    "Band: "    "EGSM"

RSRP: -113 dBm
RSRP indicates weak signal strength.
SNR: 3.5 dB
SNR indicates poor signal quality.
```

Figure 13 verification result

10.1.7)Samsung Inside of the building result

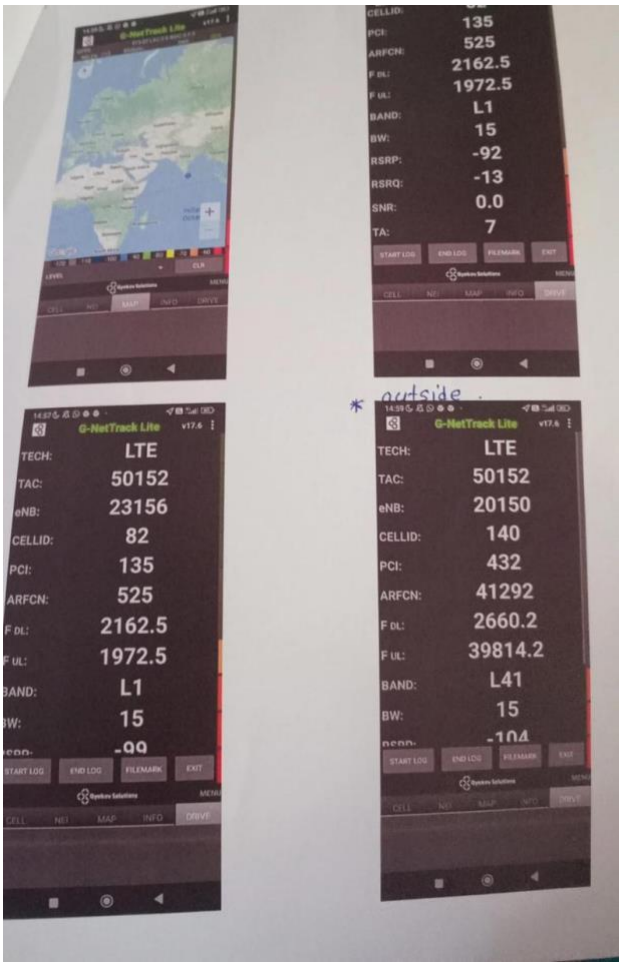


Figure 14 Samsung Inside of the building result

Band (BAND): L1. This likely indicates the LTE frequency band the phone is connected to, although the specific band isn't specified.

RSRP: -92 dBm: Indicates weak signal strength, which may result in call drops and slow data speeds.

RSRQ: -13 dBm: This value is good, suggesting a strong signal with minimal background noise.

SNR: 0.0 dBm: A very low SNR, indicating poor signal quality compared to surrounding noise. Ideally, SNR should be above 5 for optimal performance.

TECH: LTE: Confirms the phone is connected to an LTE network.

TAC: 50152: Tracking Area Code identifying a specific geographic area served by the network.

eNB: 20150: eNodeB identifier, identifying a specific cell site or base station.

CELLID: 140: Cell Identifier, unique for a particular cell within a base station.

PCI: 432: Physical Cell Identity, unique for a cell sector within a base station.

ARFCN: 41292: Absolute Radio Frequency Channel Number, identifying the carrier frequency being used.

FDL: 2660.2 MHz: Likely Frequency Downlink, the frequency range the base station uses to transmit data to the phone.

FUL: 39814.2 MHz: Likely Frequency Uplink, the frequency range the phone uses to transmit data to the base station.

BW: 15 MHz: Bandwidth, the width of the signal channel in MHz.

10.1.8)Samsung Verification result inside of the building

```
Signal Analysis:
    "Band: "    "L1"

RSRP: -92 dBm
RSRP indicates moderate signal strength.
RSRQ: -13 dB
RSRQ indicates excellent signal quality.
SNR: 0 dB
SNR indicates poor signal quality.
    "Technology: "    "LTE"

TAC: 50152
eNB: 20150
Cell ID: 140
PCI: 432
ARFCN: 41292
FDL: 2660.2 MHz
FUL: 39814.2 MHz
BW: 15 MHz
```

Figure 15 verification result

10.1.9)Redmi

10.1.10)10Redmi Inside of the building

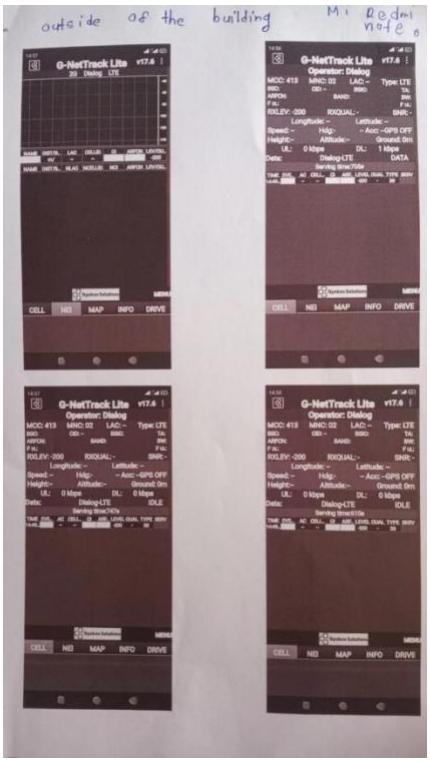


Figure 16 Redmi Inside of the building

(walk testing is not working properly)

10.1.11)Redmi Outside of the building

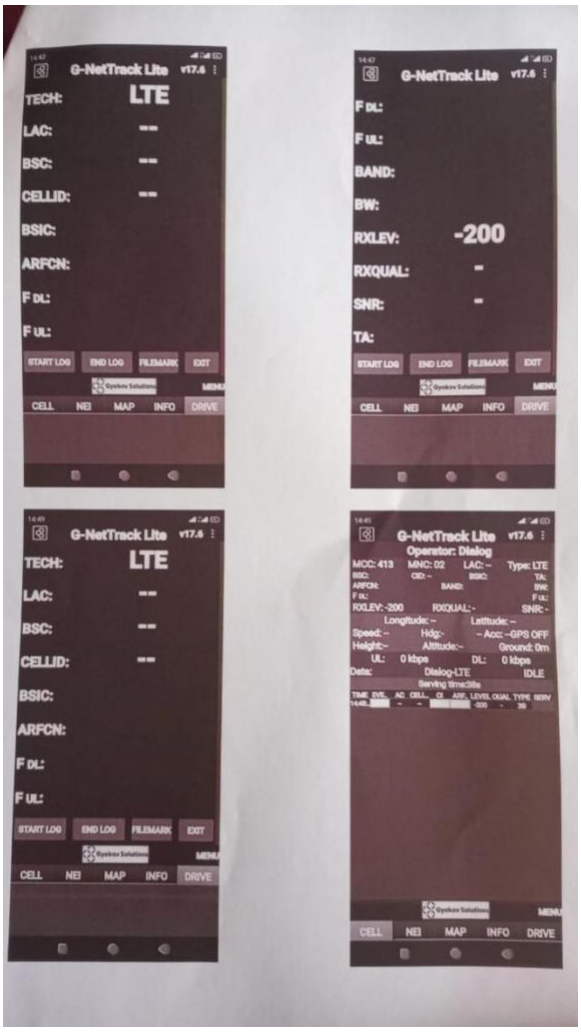


Figure 17 Redmi Outside of the building

(not working properly)

10.1.12)Galaxy J14

10.1.13)Galaxy I10nside and outside of the building

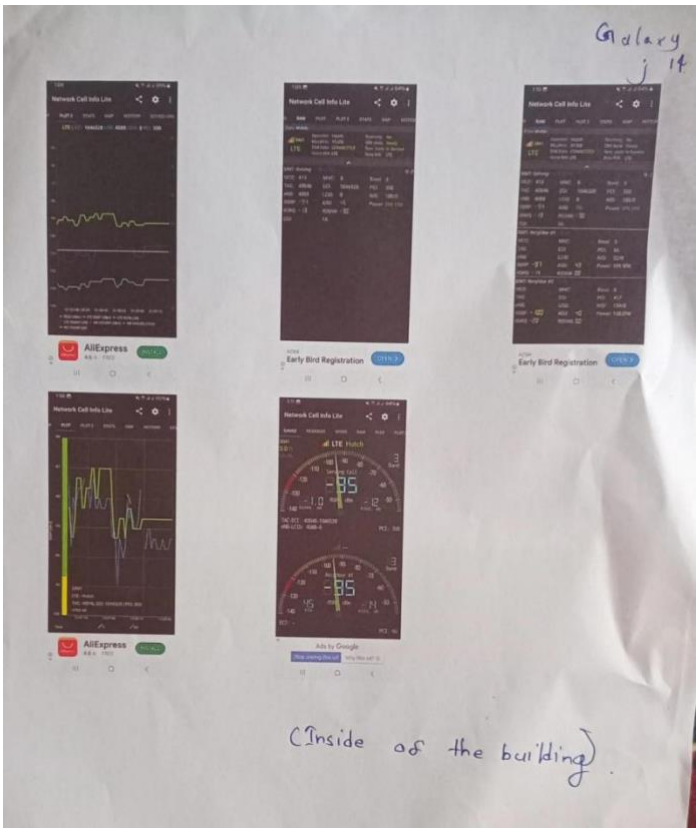


Figure 18 Galaxy I10nside and outside of the building

Band (BAND): Not available. The band information isn't displayed in this image.

RSRP: -99 dBm: Indicates weak signal strength, potentially leading to frequent call drops and slow data speeds.

RSRQ: -13 dBm: This value is good, suggesting a strong signal with minimal background noise.

SNR: 3 dBm: A low SNR, indicating poorer signal quality compared to surrounding noise. Ideally, SNR should be above 5 for optimal performance.

TECH: LTE: Confirms the phone is connected to an LTE network.

CID: 140: Likely the Cell Identifier, unique to a specific cell within a base station.

TAC: 50152: Tracking Area Code identifying a specific geographic area served by the network.

eNB: 20150: eNodeB identifier, identifying a specific cell site or base station.

PCI: 432: Physical Cell Identity, unique to a cell sector within a base station.

ARFCN: 41292: Absolute Radio Frequency Channel Number, identifying the carrier frequency being used.

FDL: 2660.2 MHz: Likely Frequency Downlink, the frequency range the base station uses to transmit data to the phone.

FUL: 39814.2 MHz: Likely Frequency Uplink, the frequency range the phone uses to transmit data to the base station.

BW: 15 MHz: Bandwidth, the width of the signal channel in MHz.

10.1.14)Galaxy j14 Verification result

```
Signal Analysis:
    "Band: "    "Not available"

RSRP: -99 dBm
RSRP indicates moderate signal strength.
RSRQ: -13 dB
RSRQ indicates excellent signal quality.
SNR: 3 dB
SNR indicates poor signal quality.
    "Technology: "    "LTE"

CID: 140
TAC: 50152
eNB: 20150
PCI: 432
ARFCN: 41292
FDL: 2660.2 MHz
FUL: 39814.2 MHz
BW: 15 MHz
```

Figure 19 Galaxy j14 Verification result

10.1.15)Result sheets of Inside the all floors using vivo phone 1 st floor

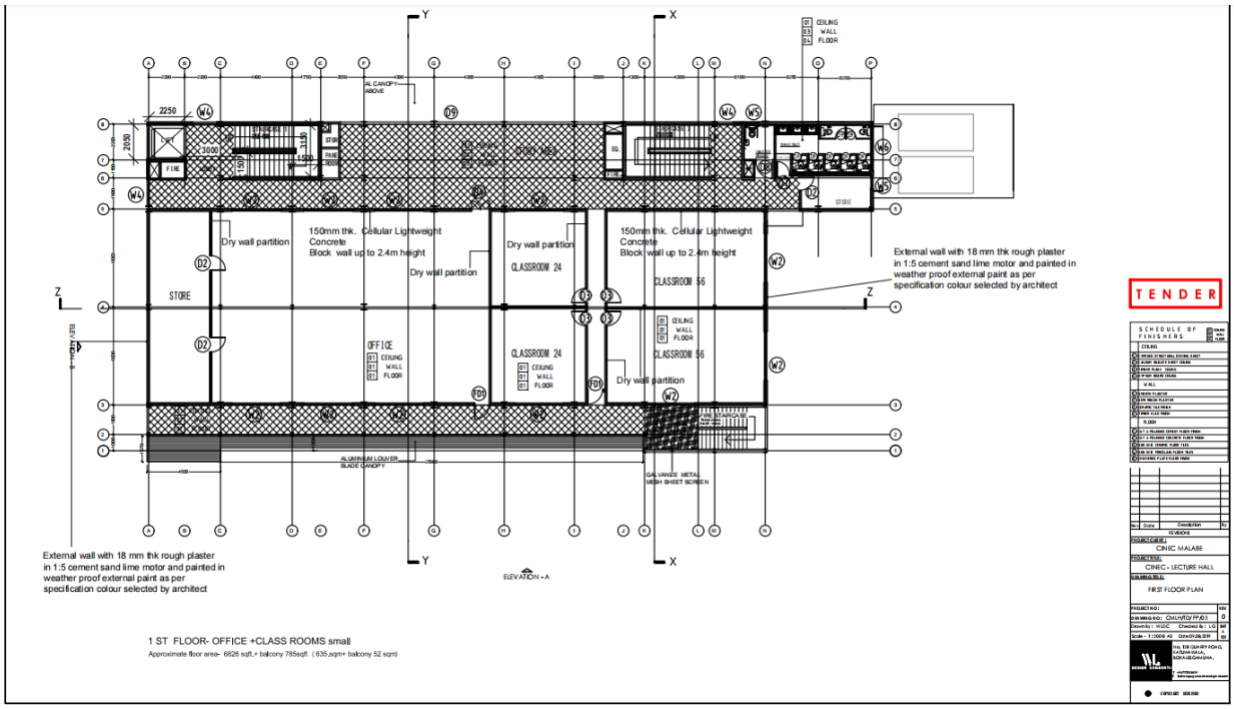


Figure 20 1 st floor cad drawing

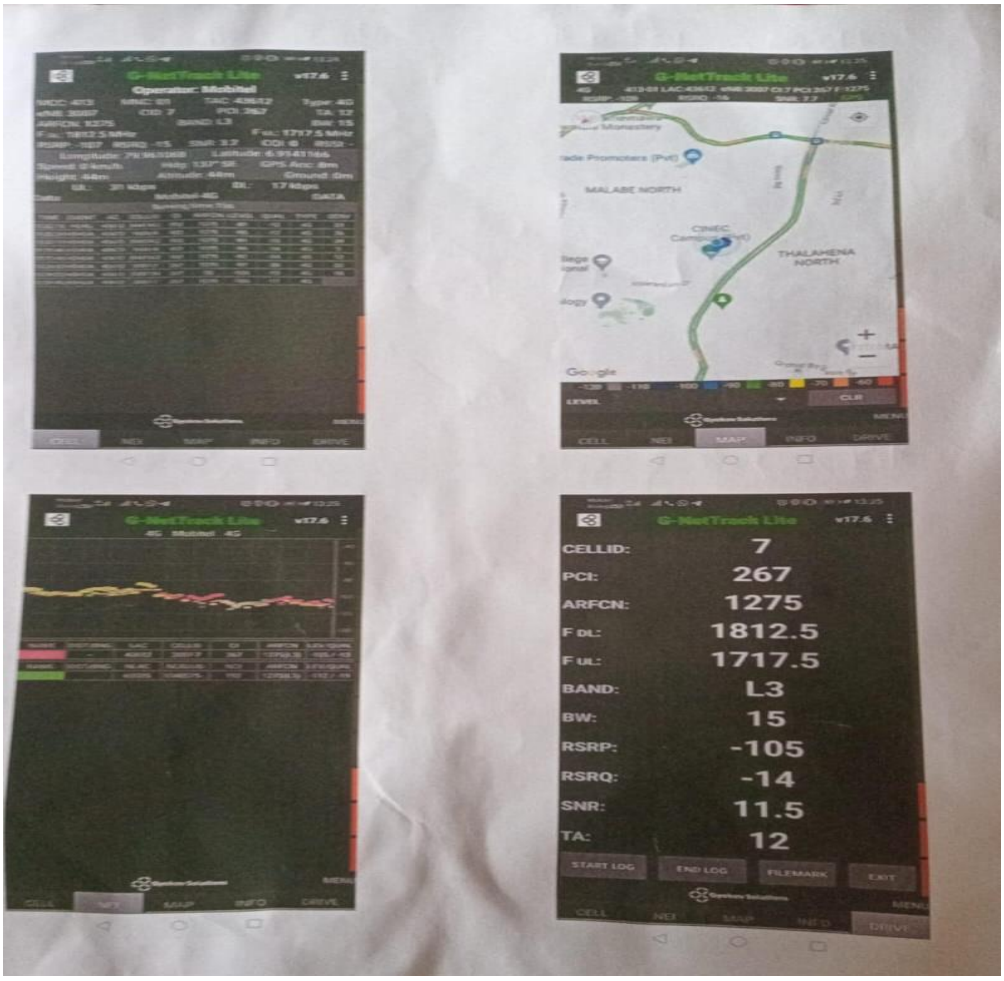


Figure 21 1st floor walk testing result

Band (BAND): Not available. The band information isn't provided in this image.

RSRP: -105 dBm: Indicates an acceptable signal strength for basic voice calls and data usage. However, for optimal indoor reception, a value between -90 dBm and -70 dBm is preferable. With -105 dBm, you might experience occasional call drops or slow data speeds.

RSRQ: -14 dBm: This value is good, indicating a strong signal with minimal background noise.

SNR: 11.5 dB: This value is good, suggesting a strong signal compared to surrounding noise. SNR values above 5 are generally considered acceptable.

CELLID: 7: Likely the Cell Identifier, unique to a specific cell within a base station.

PCE: 267: Possibly refers to Primary Cell Energy, measuring the signal strength from the cell the phone is connected to.

Signal strength seems acceptable for basic use on the 1st floor, judging by the RSRP, RSRQ, and SNR values. However, occasional call drops or slow data speeds may occur

10.1.16)Result sheets of Inside the all floors using vivo phone Verification result

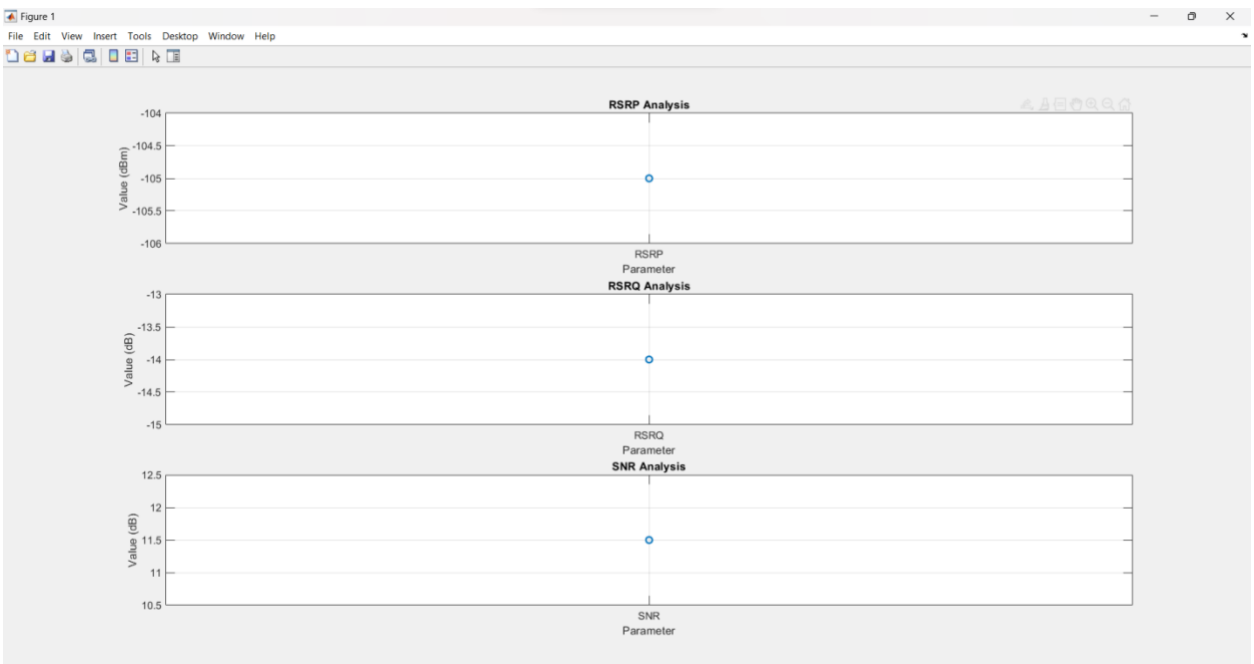


Figure 22 verification result 1st floor

```
Signal Analysis:
RSRP: -105 dBm
RSRP indicates poor indoor reception. Occasional call drops or slow data speeds may occur.
RSRQ: -14 dB
RSRQ indicates excellent signal quality.
SNR: 11.5 dB
SNR indicates good signal quality.
Cell ID: 7
PCE: 267
```

Figure 23 1 st floor verification result

10.1.17)Result sheets of Inside the all floors using vivo phone 2nd floor

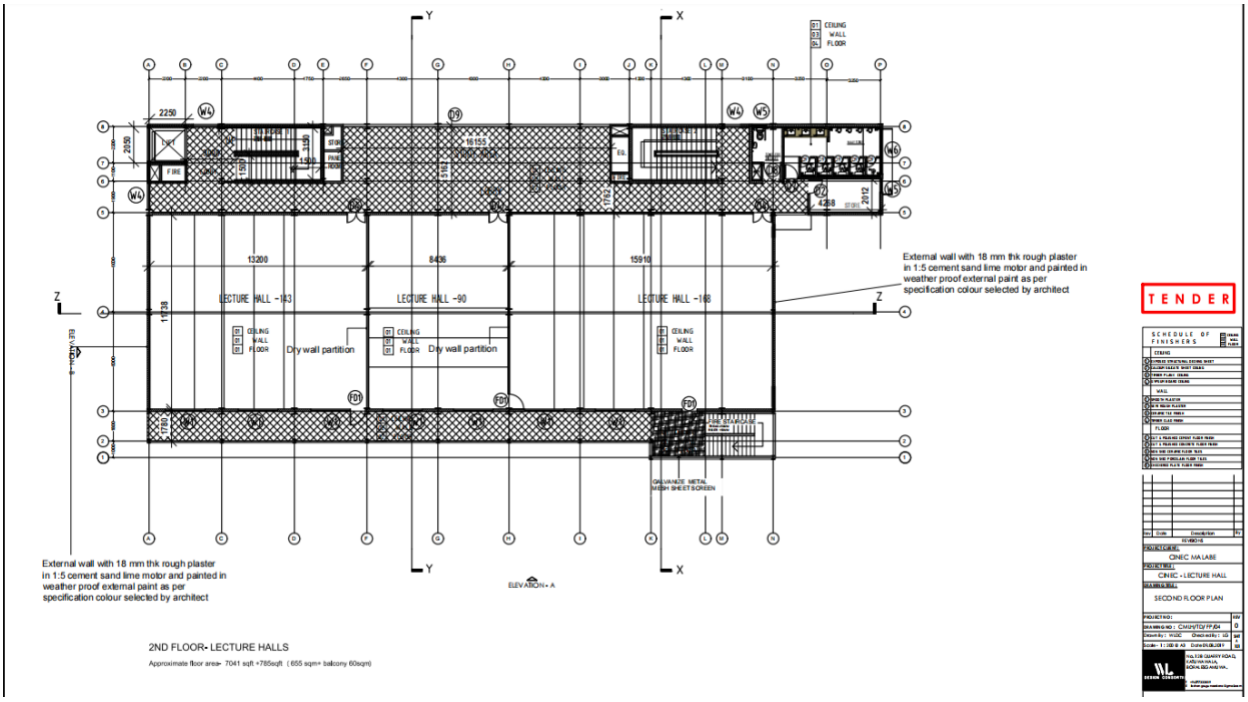


Figure 24 cad drawing of 2nd floor

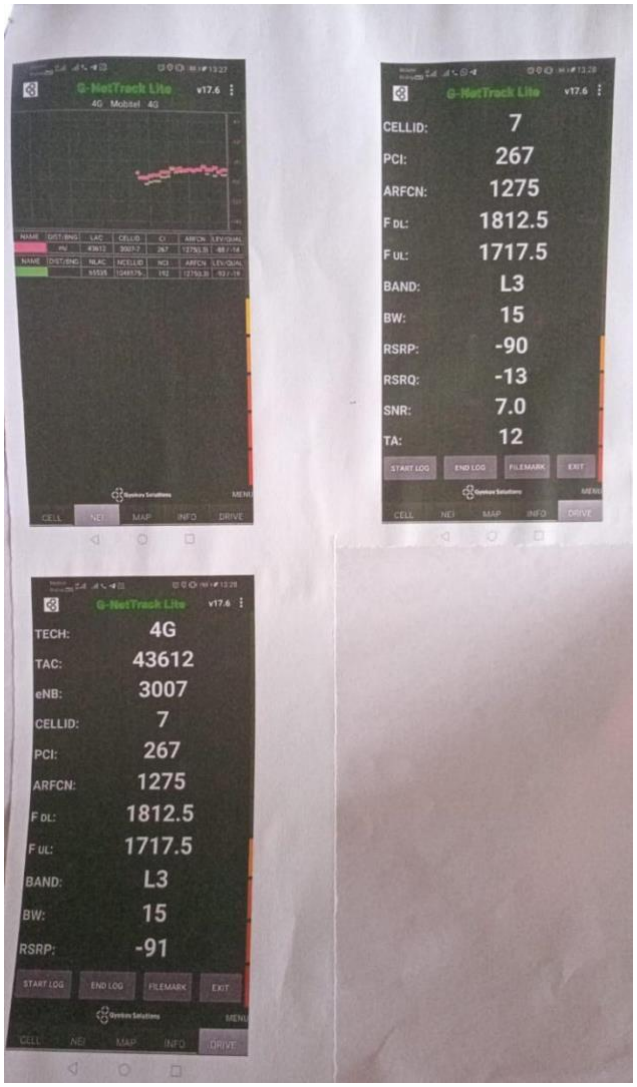


Figure 25 2nd floor walk testing result

RSRQ: -13 dBm: Falling within the lower range, -13 dBm is considered good, indicating a strong signal with minimal background noise.

BW: 15 MHz: Standard for most LTE channels, it doesn't directly indicate signal quality.

Band (BAND): L3: L3 signifies an LTE band, which is positive as LTE offers faster data speeds.

RSRP: -90 dBm: While close to acceptable for indoor reception, ideally closer to -70 dBm is preferred. -90 dBm may result in occasional call drops or slow data speeds.

SNR: 7.0 dB: Considered acceptable, with higher SNR indicating better signal quality compared to surrounding noise.

CELLID: 7: Unique identifier for a specific cell within a base station, unrelated to signal quality.

PC: 267: Likely referring to Primary Cell Energy, a relative signal strength measure within the app.

ARFCN: 1275: Identifies carrier frequency, not indicating signal quality directly.

FOL (1812.5 MHz) & FUL (1717.5 MHz): Referring to Frequency Downlink and Frequency Uplink respectively, they don't directly indicate signal quality.

BANIX: L3: Another way of indicating LTE frequency band, similar to BAND: L3, not directly related to signal quality.

TA: 12: Timing Advance value, not indicating signal quality directly.

The signal strength on the 2nd floor appears acceptable for basic use, judging by the RSRP, RSRQ, and SNR values. However, occasional call drops or slow data speeds may still occur.

.

10.1.18)Result sheets of Inside the all floors using vivo phone Verification result of 2nd floor

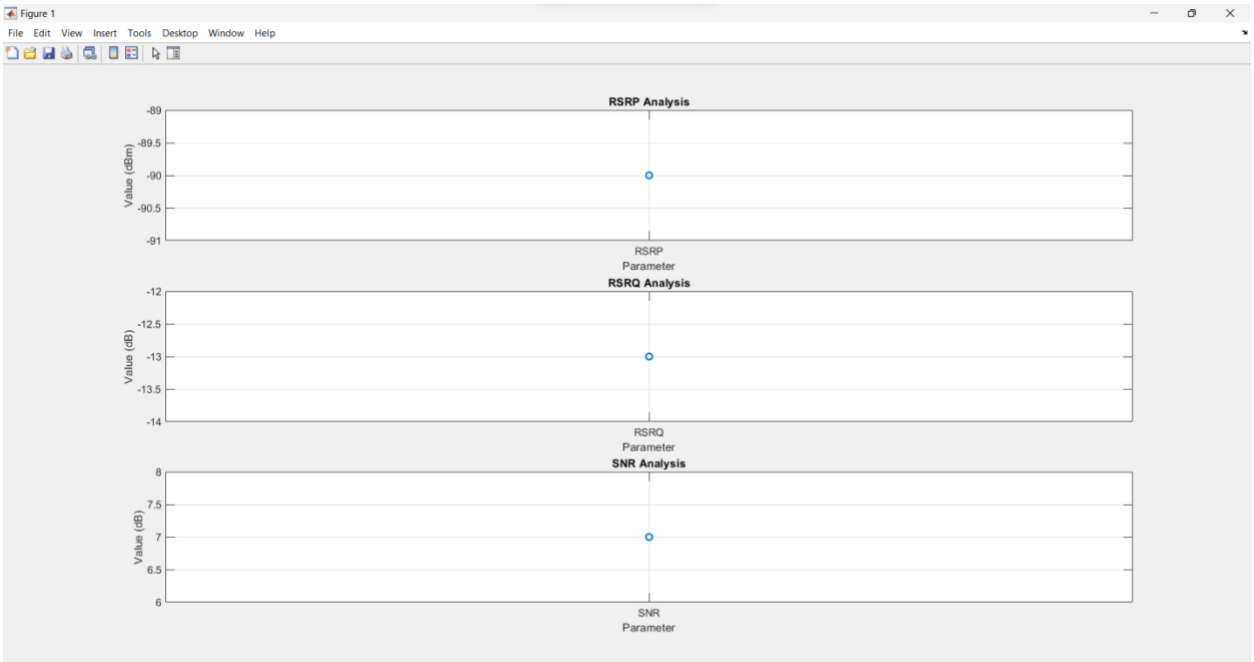


Figure 26 2nd floor verification result

```
Signal Analysis:
    "Band: "    "L3"

RSRP: -90 dBm
RSRP indicates acceptable indoor reception.
RSRQ: -13 dB
RSRQ indicates excellent signal quality.
SNR: 7 dB
SNR indicates good signal quality.
Cell ID: 7
PC: 267
ARFCN: 1275
FOL: 1812.5 MHz
FUL: 1717.5 MHz
BW: 15 MHz
TA: 12
```

Figure 27 2nd floor verification result

10.1.19)Result sheets of Inside the all floors using vivo phone 3rd floor

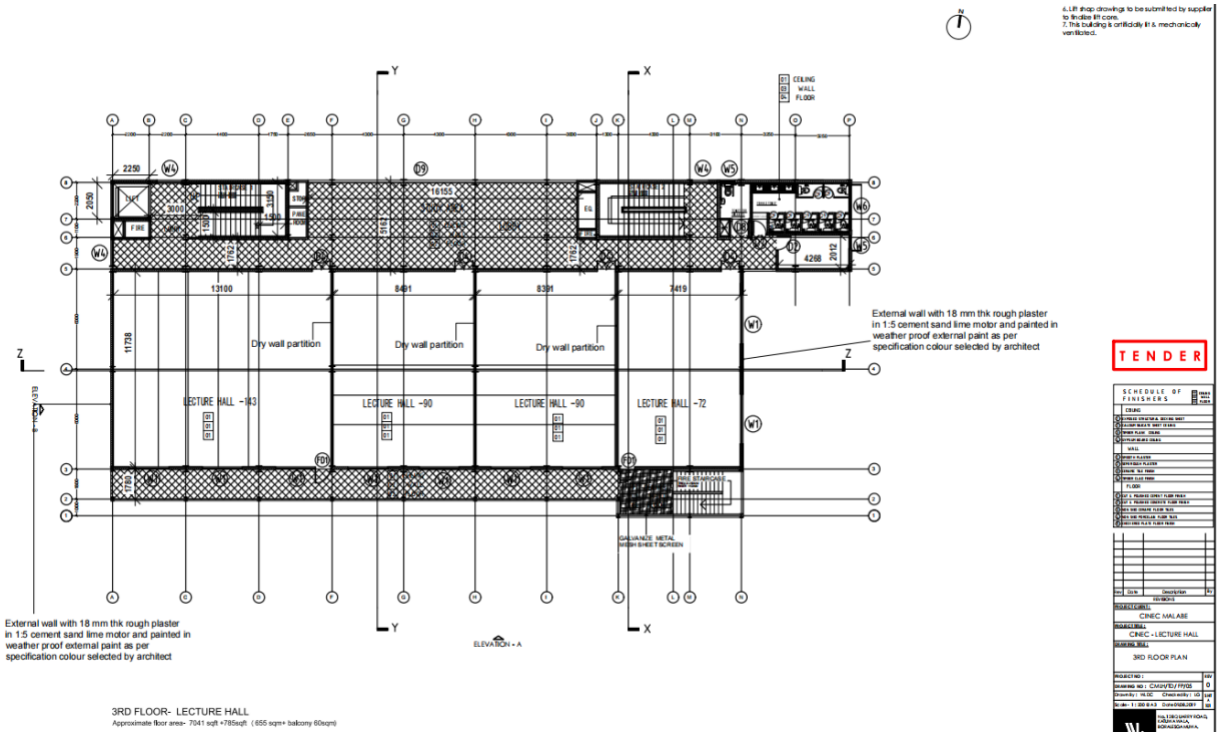


Figure 28 3rd floor cad

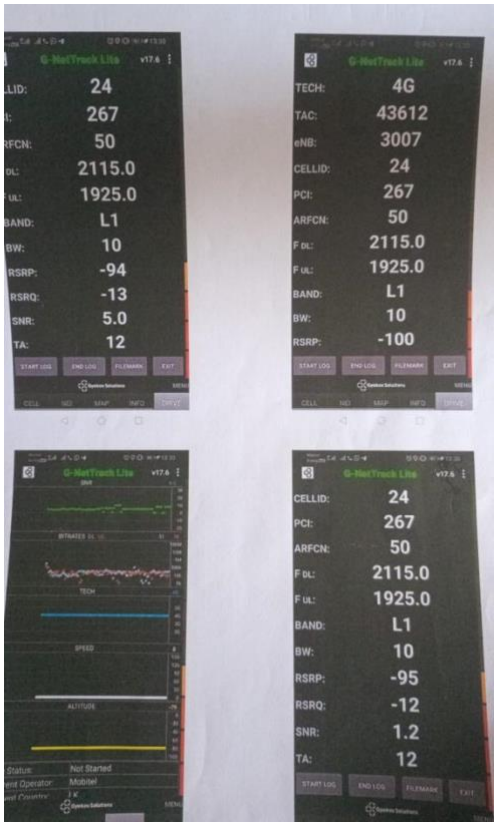


Figure 29 3rd floor walk westing result

Band (BAND): L3 - Indicates the LTE frequency band the phone is connected to. Specific band information not provided.

RSRP: -92 dBm - Acceptable signal strength for basic voice calls and data usage, though may lead to occasional call drops or slow data speeds.

RSRQ: -12 dBm - Good signal quality with minimal background noise, indicating a strong signal.

SNR: 3.5 dB - Low SNR value, suggesting poorer signal quality compared to surrounding noise. Values above 5 are preferable.

CELLID: 24 - Cell Identifier, unique to a particular cell within a base station.

PCI: 324 - Physical Cell Identity, unique identifier for a cell sector within a base station.

ARFCN: 1650 - Absolute Radio Frequency Channel Number, identifying the carrier frequency being used.

FDL: 1850.0 MHz - Likely Frequency Downlink, the frequency range the base station uses to transmit data to the phone.

FUL: 1755.0 MHz - Likely Frequency Uplink, the frequency range the phone uses to transmit data to the base station.

BW: 15 MHz - Bandwidth, the width of the signal channel in MHz.

TA: 1 - Likely Timing Advance, used to adjust the timing of the signal between the phone and the cell tower.

Signal strength seems acceptable for basic use on the 2nd floor based on the RSRP and RSRQ values. However, the SNR is low, suggesting you might experience more frequent call drops or slow data speeds compared to a higher SNR value.

10.1.20)Result sheets of Inside the all floors using vivo phone Verification result 3 rd floor

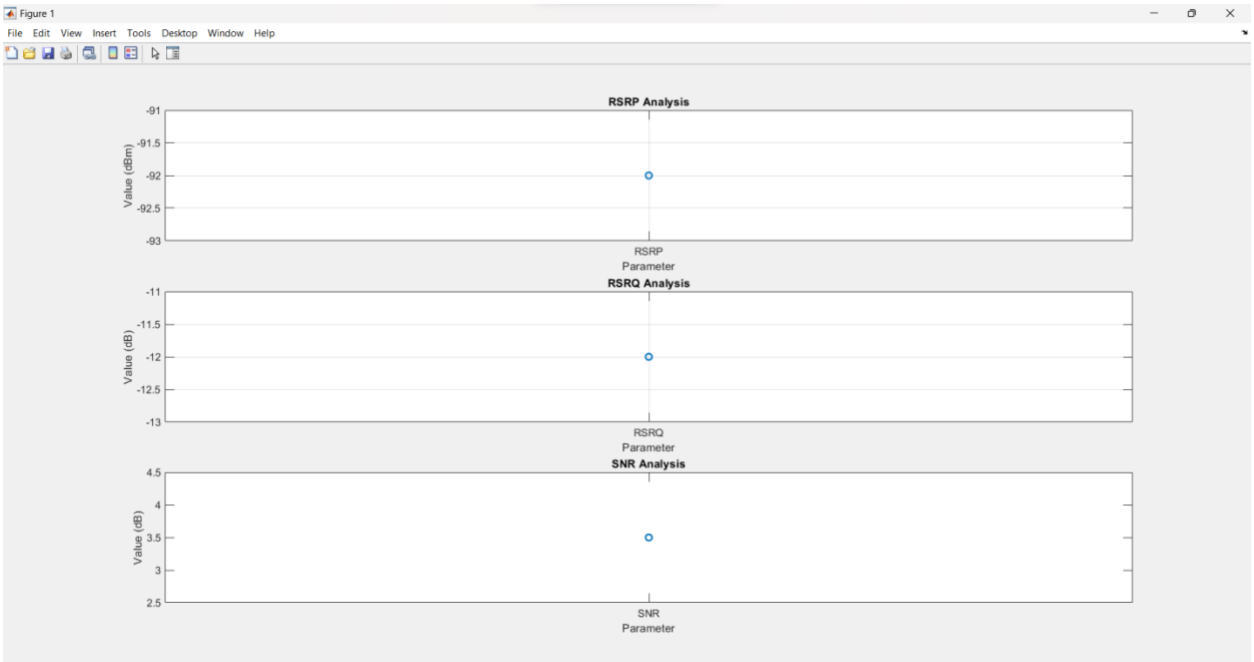


Figure 30 3rd floor verification

```
Signal Analysis:
    "Band: "    "L3"

RSRP: -92 dBm
RSRP indicates acceptable signal strength, but may lead to occasional call drops or slow data speeds.
RSRQ: -12 dB
RSRQ indicates excellent signal quality.
SNR: 3.5 dB
SNR indicates poor signal quality.
Cell ID: 24
PCI: 324
ARFCN: 1650
FDL: 1850 MHz
FUL: 1755 MHz
BW: 15 MHz
TA: 1
```

Figure 31 3rd floor verification result

10.1.21)Result sheets of Inside the all floors using vivo phone 4th floor

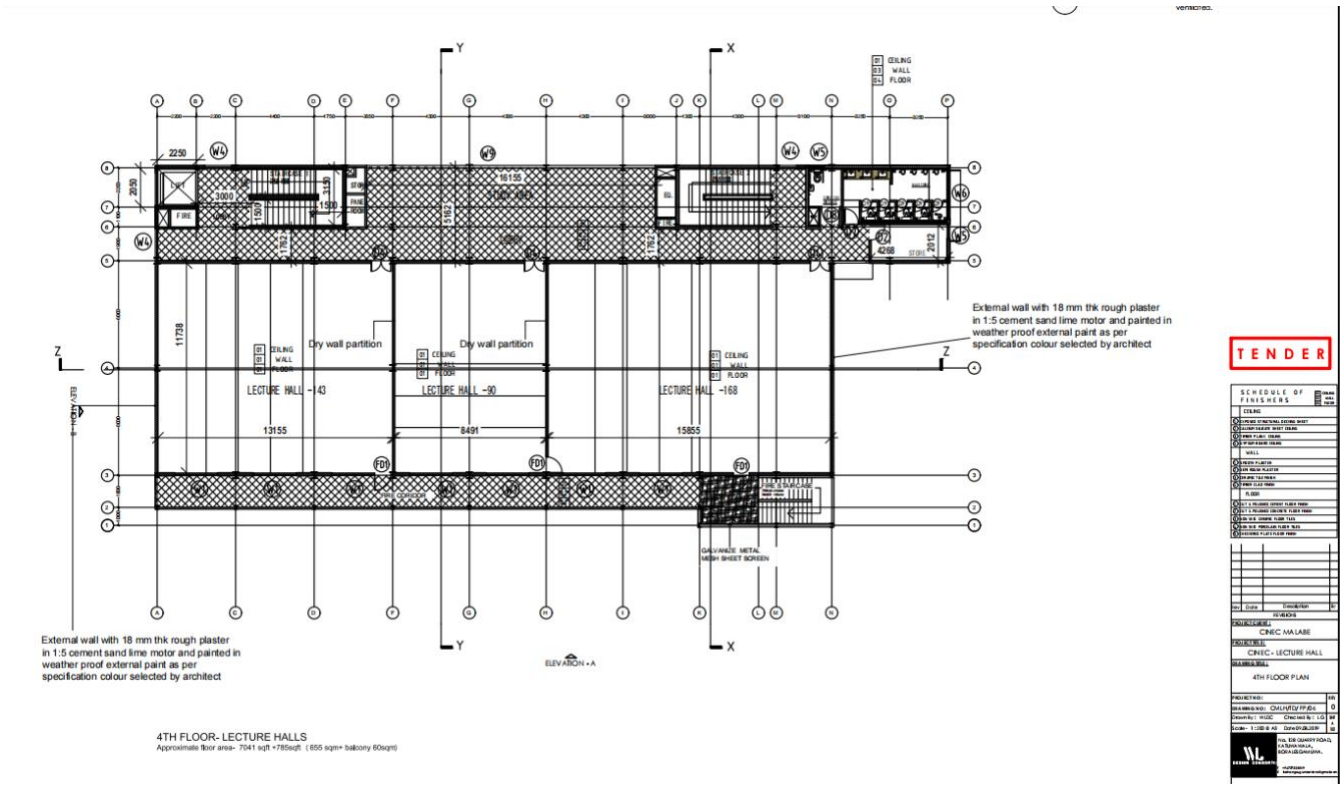


Figure 32 4th floor cad

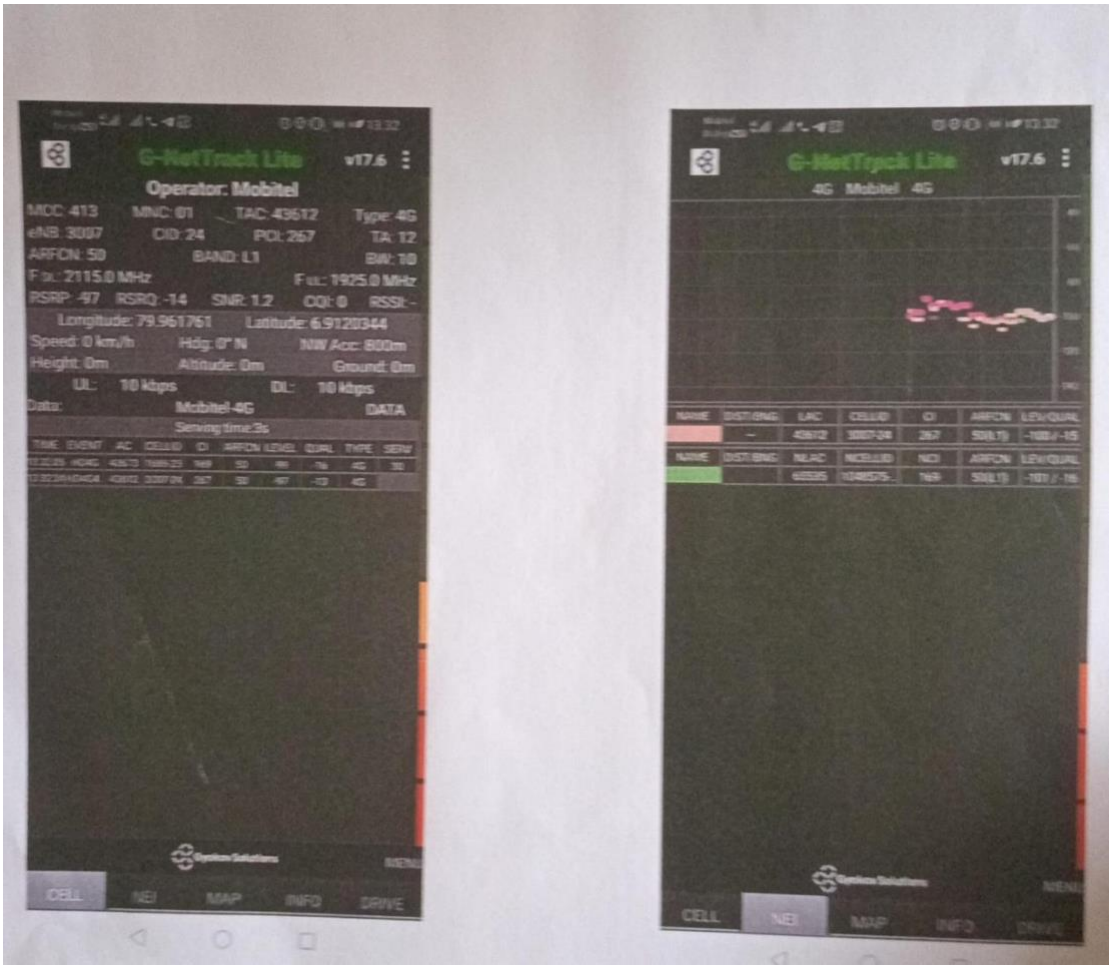
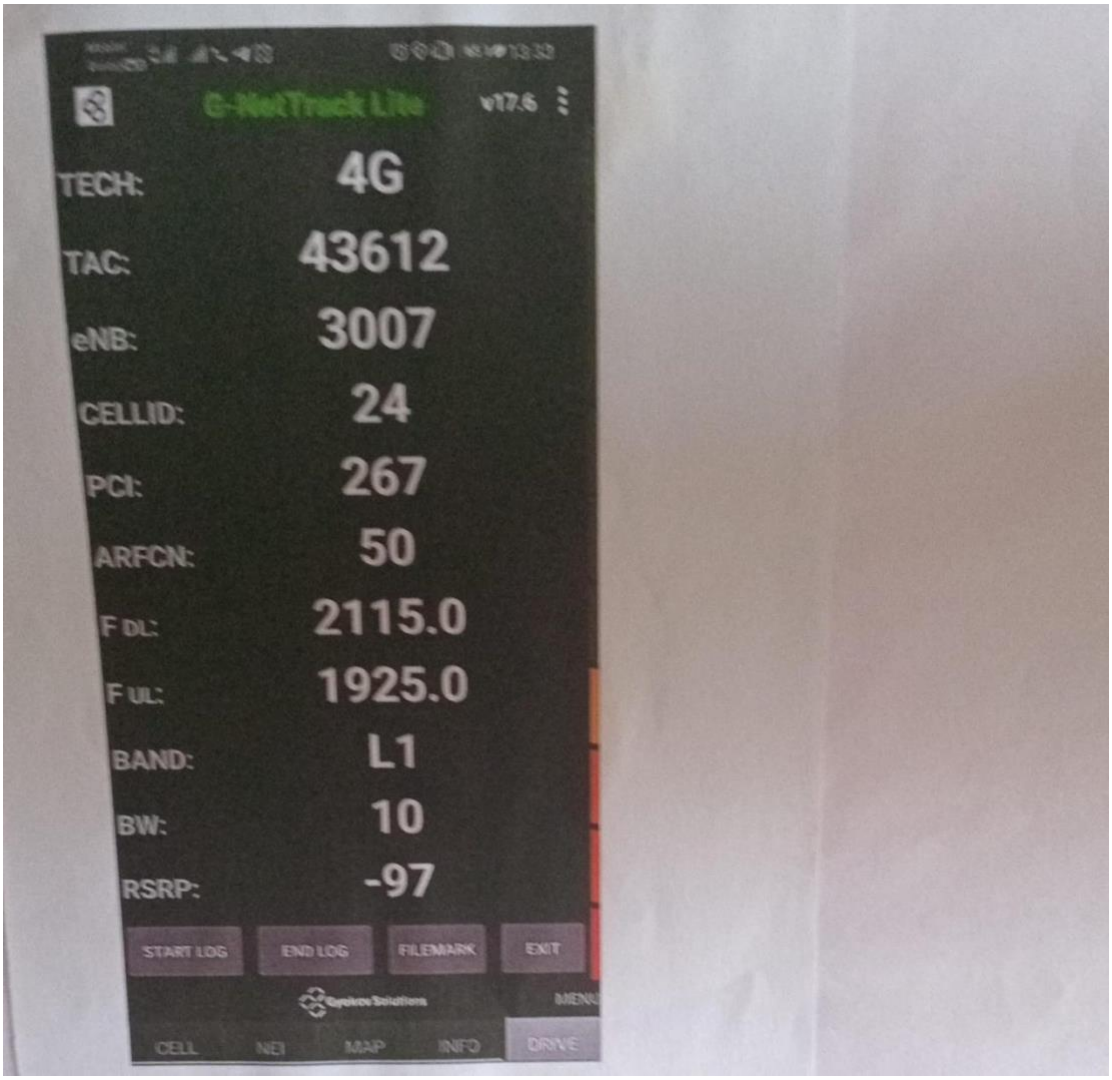


Figure 33 4th floor walk testing



Band (BAND): L1 - Likely an LTE frequency band, specific band information not provided.

RSRP: -97 dBm - Indicates poor signal strength, suggesting frequent call drops and slow data speeds.

RSRQ: -14 dBm - Good signal quality with minimal background noise, indicating a strong signal.

SNR: -2.0 dB - Very poor SNR value, suggesting significant signal degradation compared to surrounding noise.

TECH: LTE - Confirms connection to an LTE network.

TAC: 43612 - Tracking Area Code identifying a specific geographic area served by the network.

eNB: 3007 - eNodeB identifier, identifying a specific cell site or base station.

CELLID: 24 - Cell Identifier, unique to a particular cell within a base station.

PCI: 267 - Physical Cell Identity, unique identifier for a cell sector within a base station.

ARFCN: 50 - Absolute Radio Frequency Channel Number, identifying the carrier frequency being used.

FDL: 2115.0 MHz - Likely Frequency Downlink, the frequency range the base station uses to transmit data to the phone.

FUL: 1925.0 MHz - Likely Frequency Uplink, the frequency range the phone uses to transmit data to the base station.

BW: 10 MHz - Bandwidth, the width of the signal channel in MHz.

MENT: Likely an abbreviation specific to the G-NetTrack Lite app, not indicating signal strength.

Overall, the signal strength appears weak based on the RSRP and SNR values. You're likely to experience frequent call drops and slow data speeds on the 4th floor.

10.1.22)Result sheets of Inside the all floors using vivo phone Verification result 4th floor

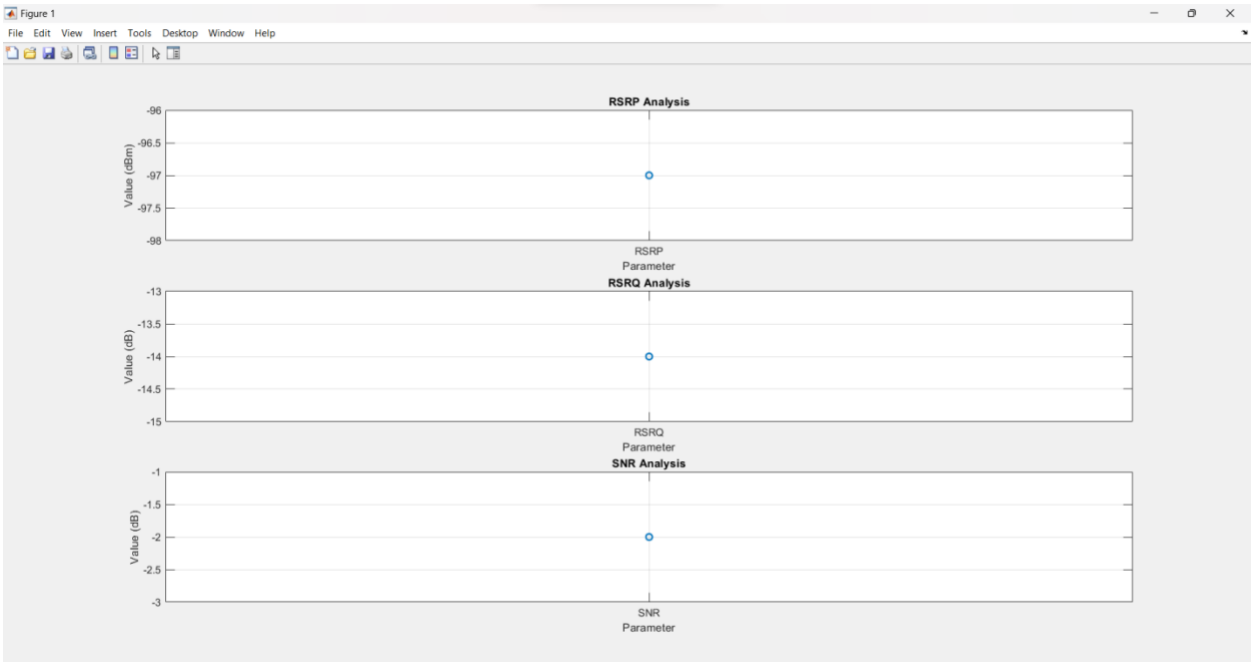


Figure 34 4th floor verification

```
Signal Analysis:
    "Band: "    "L1"

RSRP: -97 dBm
RSRP indicates poor signal strength, suggesting frequent call drops and slow data speeds.
RSRQ: -14 dB
RSRQ indicates excellent signal quality.
SNR: -2 dB
SNR indicates poor signal quality.
    "Technology: "    "LTE"

TAC: 43612
eNB: 3007
Cell ID: 24
PCI: 267
ARFCN: 50
FDL: 2115 MHz
FUL: 1925 MHz
BW: 10 MHz
```

10.1.23)Result sheets of Inside the all floors using vivo phone 5TH floor

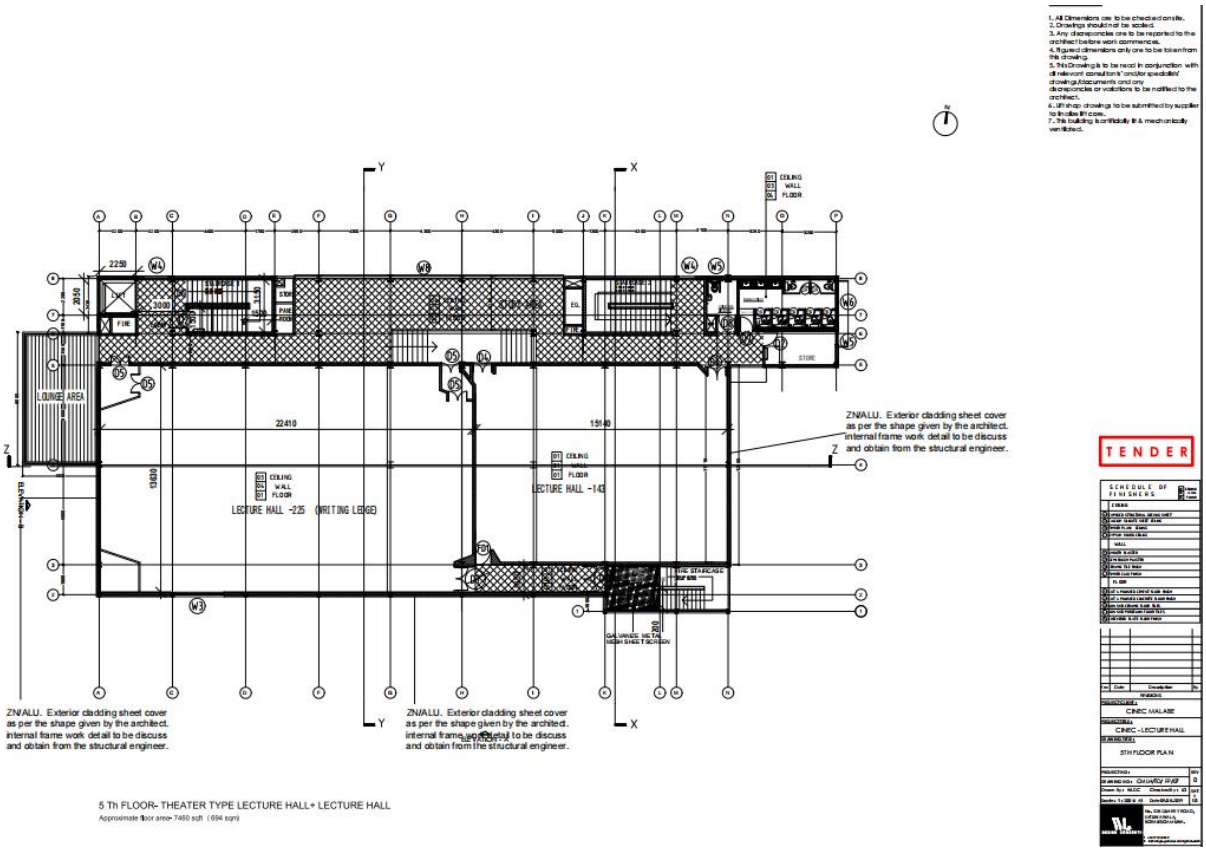


Figure 35 5th floor cad

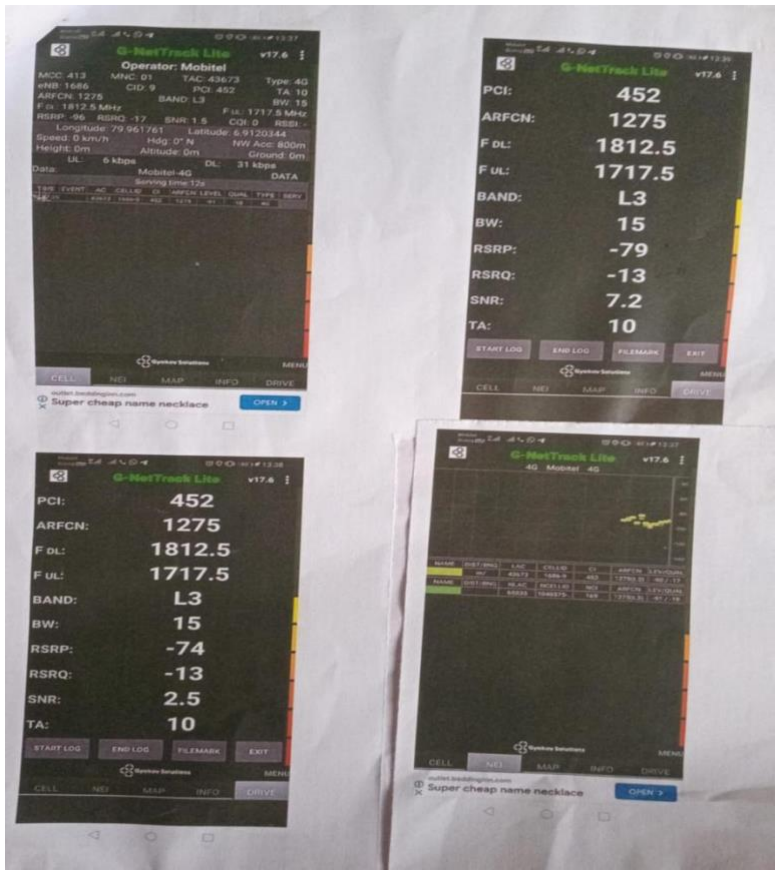


Figure 36 5th floor walk testing

Band (BAND): L3 - Likely refers to the LTE frequency band the phone is connected to, specific band information not provided.

RSRP: -74 dBm - Indicates good signal strength for indoor reception, suggesting good signal quality for calls and data usage.

RSRQ: -13 dBm - Good signal quality with minimal background noise, indicating a strong signal.

SNR: 10 dB - Good SNR value, indicating better signal compared to surrounding noise. Values above 5 are generally acceptable.

TA: 1 - Likely Timing Advance, adjusting signal timing between phone and cell tower, not directly indicating signal quality.

Other data points like CELLID, PCI, ARFCN, FDL, FUL, BW are similar to those explained previously and do not directly indicate signal strength.

Overall, the signal strength on the 5th floor appears good based on the RSRP, RSRQ, and SNR values. Experience good call quality and data speeds.

10.1.24)Result sheets of Inside the all floors using vivo phone Verification result 5th floor

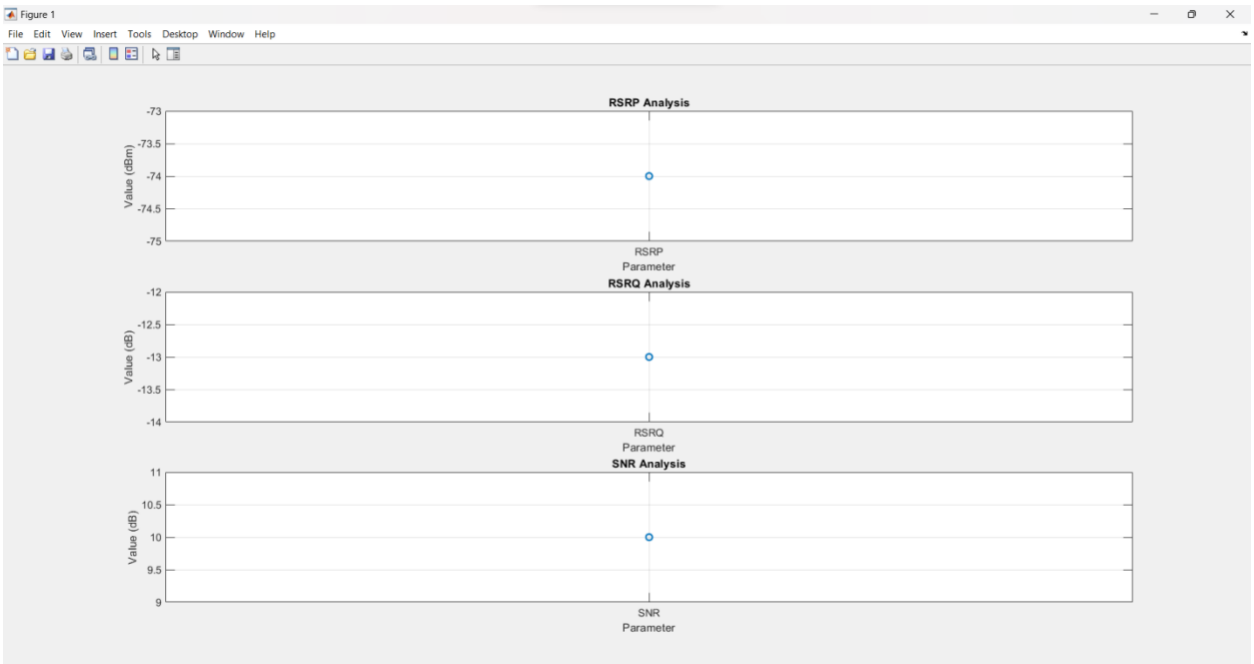
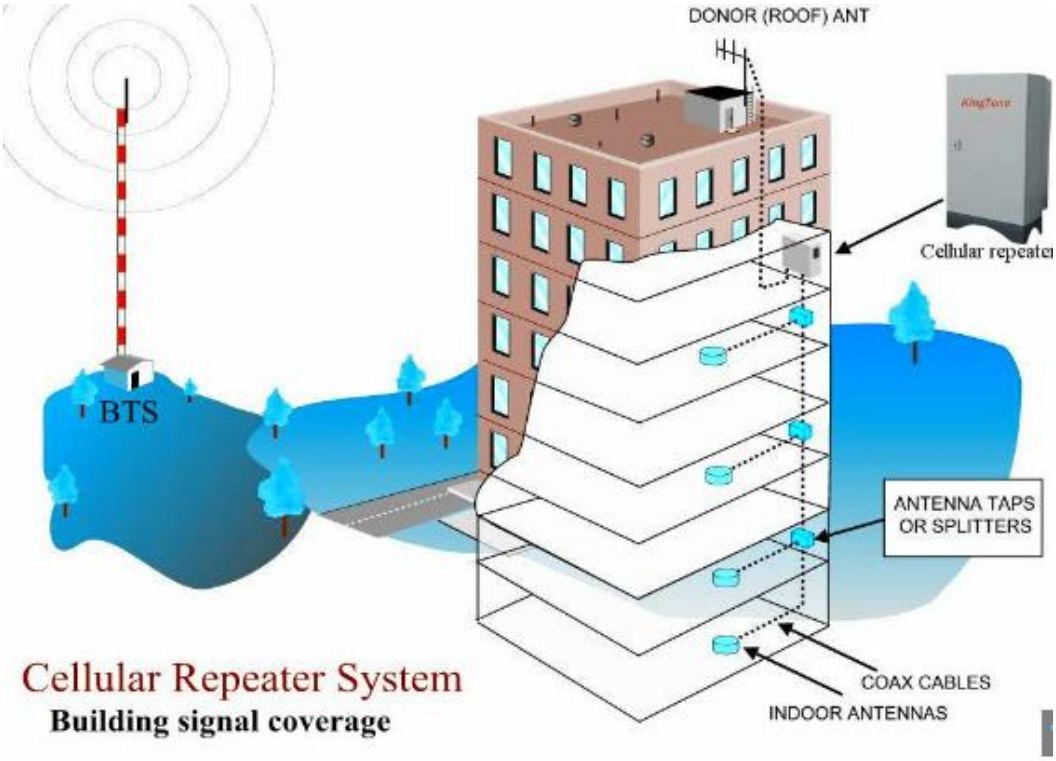


Figure 37 verification of 5th floor

```
Signal Analysis:
    "Band: "    "L3"

RSRP: -74 dBm
RSRP indicates excellent signal strength for indoor reception.
RSRQ: -13 dB
RSRQ indicates excellent signal quality.
SNR: 10 dB
SNR indicates good signal quality.
TA: 1
```

10.1.25)Solution- Install repeater system (IBS)



Choosing Donor Antennas for a 7-Floor Building with 2G and 4G Bands (8350 sq ft per Floor)

Building Parameters:

- Number of Floors: 7
- Coverage Area per Floor: 777 sqm (8350 sqft)
- Frequency Bands: 2G and 4G

Donor Antenna Selection:

Following factors are crucial when choosing donor antennas for this ,

Frequency Coverage: Select antennas compatible with both 2G and 4G bands. Look for antennas with wideband or multiband designations.Coverage Pattern: Considering the square-shaped floor plan, omni-directional antennas are suitable for radiating signals in all directions.

Gain: Higher gain antennas provide better signal strength over longer distances, but they might not be ideal for even indoor coverage within a limited area like 8350 sqft. Medium gain antennas (around 6-8 dBi) offer a good balance between coverage and signal strength.

Ceiling Penetration: Depending on the ceiling material (e.g., concrete, drywall), consider antennas with good ceiling penetration capabilities to ensure sufficient signal reaches lower floors.

Beamtilt: Slight downward tilt (5-10 degrees) can be beneficial to focus the signal coverage area more towards the floor space, improving indoor reception.

Sample Donor Antenna Recommendation-

Based on the above criteria, a possible donor antenna option could be:

- Medium Gain Omni-directional Antenna (6-8 dBi gain)

- Multiband Operation (covering 2G and 4G bands)
- Ceiling-mount type with appropriate tilt angle
- Number of Donor Antennas per Floor Calculation:

Here, we can estimate the number of antennas needed using a simplified path loss model and considering a minimum acceptable signal strength threshold. However, this is a theoretical approach, and a real-world deployment might require adjustments based on a site survey and radio planning software.

Assumptions:

Minimum acceptable signal strength: -70 dBm (considering some margin)

Building material: Drywall (assuming moderate penetration loss)

Frequency band: We'll calculate for 850 MHz (2G) as it typically has higher path loss compared to 4G bands.

Path Loss Calculation:

Use the free space path loss equation at a reference distance (1 meter):

Free Space Path Loss (dB) = 32.45 + 20 * log10(Frequency in MHz) For 850 MHz: Path Loss = 32.45 + 20 * log10(850) = 109.45 dB

Maximum Distance Calculation

We want the signal strength to be at least -70 dBm at the coverage edge (considering a square floor plan, distance from the antenna to the farthest corner is the radius of the coverage circle).

Use the following equation to find the maximum distance (radius) where the signal level reaches -70 dBm: Path Loss (dB) = Tx Power (dBm) - Received Signal Strength (dBm)

Assuming a typical donor antenna Tx power of 20 dBm: 109.45 dB = 20 dBm - (-70 dBm) Maximum distance = 99.45 meters (radius)

Number of Antennas-

Calculate the area covered by a single antenna assuming a circular coverage pattern,Area per Antenna = pi * (Radius)^2

Area per Antenna = pi * (99.45 meters)^2 = 31,168 sqm

Divide the total floor area by the area covered by one antenna to estimate the number of antennas required:

Number of Antennas = Total Floor Area / Area per Antenna

Number of Antennas = 8350 sqft (converted to sqm) / 31,168 sqm/antenna

Number of Antennas (rounded up) = 1 antenna (per floor)

This calculation provides an initial estimate. In practice, a site survey and radio planning software are recommended to account for factors like wall penetration loss, actual antenna patterns, and building materials. Additionally, using 2 antennas per floor with a lower gain might provide better signal distribution and redundancy.

Based on the simplified calculations, a medium gain (6-8 dBi) omni-directional multiband antenna with a slight downward tilt could be a suitable choice for each floor. However, a real-world deployment would likely require a more comprehensive analysis using site-specific data and radio planning tools.

10.1.26)Cellular repeater

Choosing a Cellular Repeater for a 7-Floor Building (8350 sq ft per Floor) with 2G and 4G Bands

Cellular repeaters amplify existing cellular signals to improve coverage and signal strength within a building. Here's a guide to choosing a suitable repeater for your scenario:

Factors to Consider:

Frequency Bands: Ensure the repeater supports both 2G and 4G bands you require (e.g., 850 MHz for 2G, 700 MHz or higher for 4G depending on your carrier).

Gain: Higher gain repeaters amplify signals more but might introduce interference. Choose a gain suitable for your building size.

Low Gain (up to 65 dB): Suitable for smaller areas or situations where strong donor signal exists outside.

Medium Gain (65-75 dB): Good balance for moderate-sized buildings with moderate signal strength outside.

High Gain (75 dB and above): Ideal for large buildings or areas with very weak outdoor signals, but requires careful planning to avoid interference.

Coverage Area: Consider the repeater's coverage specifications and how many floors it can effectively cover.

Signal Strength Requirements: Determine the minimum acceptable signal strength for good call quality (around -70 dBm is a common target).

Donor Signal Strength: The repeater amplifies an existing outdoor signal. Choose a location with a good, stable signal from the cell tower.

Cable Types and Lengths: Select cables compatible with the chosen frequency bands and ensure they are long enough to reach the donor antenna and indoor antennas.

Sample Cellular Repeater Recommendation:

Given the 7-floor building and 8350 sq ft per floor, a possible repeater option could be,Bi-directional Repeater (amplifies both uplink and downlink signals) supporting 2G and 4G bands.Medium Gain (around 65-70 dB) to balance amplification and potential interferenceCoverage area suitable for at least 2-3 floors (might require multiple repeaters depending on final signal strength measurements)

Number of Repeaters,The exact number of repeaters depends on several factors, includings-

Measured donor signal strength

Desired signal strength throughout the building

Cable limitations (maximum cable length)

A site survey to measure signal strength at different locations is crucial for determining the optimal repeater placement and quantity.

Additional Considerations:

Installation: Cellular repeater installation might require permits and professional expertise to ensure proper configuration and compliance with regulations.

Interference: High gain repeaters can cause interference if not installed correctly. A qualified installer can minimize this risk.

Alternatives: Depending on the building layout and signal strength, a Distributed Antenna System (DAS) might be a more scalable solution for larger buildings or uneven signal distribution.

While a medium gain bi-directional repeater supporting 2G and 4G bands is a possible starting point, a site survey and professional consultation are highly recommended for optimal results. This ensures you choose the right repeater with the appropriate gain, placement, and configuration to achieve good cellular coverage throughout your building.

10.1.27)Spliters and antenna selections

Choosing Splitters and Antennas for Indoor Cellular Coverage (2G & 4G)

Understanding Signal Distribution:

In a cellular signal amplification system for a 7-floor building with 2G and 4G bands (like a cellular repeater system), strategically placed antennas and splitters are essential for distributing the amplified signal effectively. Here's a breakdown of how to choose them:

Splitters:

Function: Splitters divide the signal from a single source (amplified signal from the repeater) into multiple outputs to feed various indoor antennas.

Types:

2-way Splitters: Divide the signal into two equal outputs (suitable for feeding two antennas on the same floor).

3-way Splitters: Divide the signal into three equal outputs (useful for covering three separate areas or floors).

Higher-way Splitters: Available for distributing to more antennas, but signal strength weakens with each split (use them cautiously).

Selection Considerations-

Frequency Range: Ensure the splitter supports the 2G and 4G bands you're using.

Insertion Loss: This specifies the signal strength reduction caused by the splitter. Choose splitters with low insertion loss (ideally below 3 dB) to minimize signal degradation.

Number of Outputs: Select a splitter with enough outputs to feed the desired number of antennas.

Directional Antennas-

Function: Focus and direct the amplified signal in a specific direction, ideal for targeting weak signal areas or avoiding obstacles.

Types:

Panel Antennas: Flat, rectangular antennas with a directional radiation pattern, suitable for corridors or specific coverage zones.Yagi Antennas: Highly directional antennas with a narrow beam, useful for long distances or penetrating through walls.

Selection Considerations-

Frequency Range: Similar to splitters, ensure the antenna supports both 2G and 4G bands.

Gain: Higher gain antennas provide stronger signal in the targeted direction, but lower gain offers wider coverage. Choose based on your specific needs (focusing weak areas vs. general floor coverage).

Beamwidth: This specifies the angle of the directional pattern. Narrower beams provide more focused coverage, while wider beams offer a larger coverage area.

Omni-directional Antennas-

Function: Radiate signals in all horizontal directions, providing even coverage within a circular area.

Types:

Dipole Antennas: Simple, rod-shaped antennas with a donut-shaped radiation pattern, good for central locations on a floor.

Ceiling-mount Antennas: Designed for ceiling installation, offering good omnidirectional coverage within a floor.

Selection Considerations-

Frequency Range: Same as above, ensure compatibility with 2G and 4G bands.

Gain: Medium gain antennas provide a balance between coverage area and signal strength. Choosing Splitters and Antennas Together:

Matching Needs: Consider the building layout, desired coverage pattern, and signal strength requirements.

Start with Omni-directional Antennas- For general even coverage on each floor, start with omni-directional antennas like ceiling-mount types.Directional Antennas for Specific Areas: If there are weak signal zones or areas requiring focused coverage (e.g., hallways), add directional antennas strategically.Splitter Selection Based on Antenna Count: Choose a splitter with enough outputs to feed all the antennas on a particular floor.

Example Scenario:

For a 7-floor building with moderate signal strength variation, you might consider.Omni-directional ceiling-mount antennas on each floor for general coverage.2-way splitters to feed two omni-directional antennas per floor.Directional panel antennas in specific locations (e.g., hallways) to address weak signal areas.Choosing Splitters and Antennas for Indoor Cellular Coverage (2G & 4G)

Understanding Signal Distribution-

In a cellular signal amplification system for a 7-floor building with 2G and 4G bands (like a cellular repeater system), strategically placed antennas and splitters are essential for distributing the amplified signal effectively. Here's a breakdown of how to choose them:

Splitters:

Function: Splitters divide the signal from a single source (amplified signal from the repeater) into multiple outputs to feed various indoor antennas.

Types:

2-way Splitters: Divide the signal into two equal outputs (suitable for feeding two antennas on the same floor).

3-way Splitters: Divide the signal into three equal outputs (useful for covering three separate areas or floors).

Higher-way Splitters: Available for distributing to more antennas, but signal strength weakens with each split .

Selection Considerations-

Frequency Range: Ensure the splitter supports the 2G and 4G bands you're using.

Insertion Loss: This specifies the signal strength reduction caused by the splitter. Choose splitters with low insertion loss (ideally below 3 dB) to minimize signal degradation.

Number of Outputs: Select a splitter with enough outputs to feed the desired number of antennas.

Directional Antennas:

Function: Focus and direct the amplified signal in a specific direction, ideal for targeting weak signal areas or avoiding obstacles.

Types-

Panel Antennas: Flat, rectangular antennas with a directional radiation pattern, suitable for corridors or specific coverage zones.

Yagi Antennas: Highly directional antennas with a narrow beam, useful for long distances or penetrating through walls.

Selection Considerations-

Frequency Range: Similar to splitters, ensure the antenna supports both 2G and 4G bands.

Gain: Higher gain antennas provide stronger signal in the targeted direction, but lower gain offers wider coverage. Choose based on your specific needs (focusing weak areas vs. general floor coverage).

Beamwidth: This specifies the angle of the directional pattern. Narrower beams provide more focused coverage, while wider beams offer a larger coverage area.

Omni-directional Antennas:

Function: Radiate signals in all horizontal directions, providing even coverage within a circular area.

Types-

Dipole Antennas: Simple, rod-shaped antennas with a donut-shaped radiation pattern, good for central locations on a floor.

Ceiling-mount Antennas: Designed for ceiling installation, offering good omnidirectional coverage within a floor.

Selection Considerations-

Frequency Range: Same as above, ensure compatibility with 2G and 4G bands.

Gain: Medium gain antennas provide a balance between coverage area and signal strength.

Choosing Splitters and Antennas Together

Matching Needs: Consider the building layout, desired coverage pattern, and signal strength requirements.

Start with Omni-directional Antennas: For general even coverage on each floor, start with omni-directional antennas like ceiling-mount types.

Directional Antennas for Specific Areas: If there are weak signal zones or areas requiring focused coverage (e.g., hallways), add directional antennas strategically.

Splitter Selection Based on Antenna Count: Choose a splitter with enough outputs to feed all the antennas on a particular floor.

Example -

For a 7-floor building with moderate signal strength variation, you might consider

Omni-directional ceiling-mount antennas on each floor for general coverage.

2-way splitters to feed two omni-directional antennas per floor.

Directional panel antennas in specific locations (e.g., hallways) to address weak signal areas.

11.0)CONCLUTION

The project focused on establishing and validating a cellular mobile reception indoor terrain model for the Zenith building situated within the campus. Through meticulous data collection, analysis, and model development, the team acquired significant insights into the intricacies of cellular signal propagation within the building's confines. The findings highlight the critical importance of accurately modeling indoor terrain to optimize cellular coverage and improve user experience. By implementing the developed model, the project aims to address areas of weak reception and enhance connectivity within the Zenith building, offering benefits to both occupants and stakeholders. This endeavor underscores the team's dedication to advancing in-building communication systems while ensuring efficient and reliable cellular services within the campus infrastructure.

12.0)BIBLIOGRAPHY

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13.0)PCS AND SOA SLIDES

Individual Research Project (BSc Batch 03)

Initial Presentation

K.K.Jayani Nadeera
F19990514008

Project Conception Slide (PCS)

Project Details

Current gap: Non availability of a propagation model for buildings in Sri Lanka and non-availability of studies done for outdoor to indoor propagation of cellular signal
Project area / topic:Electronic and Telecommunication Engineering/Determination & Verification of a cellular mobile reception indoor terrain model for Zenith Building, CINEC Campus
Novelty: The proposed project will be for steel structure plus masonry and concrete fill of the zenith building in Malabe CINEC Campus and for Outdoor to indoor propagation
Grant / company / source of fund:
Duration: 8 months

SWOT and current capabilities

Strength: Local Knowledge, Potential for Impact Sustainability Goals.
Weakness: Resource Constraints, technical challenges
Opportunities: Technology advancements
Threats: Environmental factors

Current Capabilities: Technical competence and Financial resources

BIP

Expected deliverables - After determination of indoor model for the building with floor material, wall material, structure composition, furniture, celling fixtures, age etc as input parameters, the findings can be extended to any location of the building and possibly to a different building of a different structure.

Benefited industry clusters - Telecommunication Industry

TRL - TRL 3

FIP -

Team: PL, collaborator and team

Budget

	Incremental (cash)	In-kind
Eqpt	$100(including travelling & software costs)	xxxxx
EOM	$46.2	15 HOURS(1000 Phr
OOE	xxxx	

Invention	Inventors	Advantages	Disadvantages	Status of Invention	Novelty of Proposed project against those	Suggested for further development	Remarks	Refs
Propagation Measurements and Models for Wireless Communications Channels	J0rgen Bach Andersen, Theodore 5. Rappaport, and Susumu Yoshida	Physics of propagation, outdoor propagation, RF penetration into buildings are explained. Computer aided design for In-building propagation prediction, Propagation between floors, Propagation inside single-storey buildings etc are also explained.	The work had been publicized in 1995. It is not specifically for indoor environment.	The authors are reputed in Cellular Communications field. Much investigations remain to be done.	Novel ways of classifying wireless environment. These novel ways will be used in the proposed project..	3D numerical modelling and consideration of angle of arrival is required.		[1]
Prediction of Outdoor and Outdoor-to-Indoor Coverage in Urban Areas at 1.8 GHz	Thomas Kürner, Senior Member, IEEE and Alexander Meier	The proposed model is efficient and accurate, and can predict indoor coverage.	Data requirements: The model relies on high-resolution building data, which may not be readily available in all regions.	The radio propagation prediction model is still in its early stages of development, but it has shown promising results in preliminary tests.	Integrated Prediction Model: The authors have developed an integrated three-dimensional radio propagation prediction model that combines outdoor and outdoor-to-indoor coverage predictions within one framework. These novel ways will be used in the proposed project	Validate the model in real-world environments.	The model has the potential to be integrated into a broader outdoor prediction model, enhancing cellular mobile system design.	2] J. D. Foley et al., Computer Graphics—Principles and Practice. Reading, MA: Addison-Wesley, 1993.

Measurement of Building Penetration Loss and Propagation Models for Radio Transmission into Buildings	R. Hoppe, G. Wolfle, and F. M. Landstorfer	Improved Indoor Coverage: Propagation models for radio transmission into buildings enhance the ability to calculate indoor field strength coverage based on outdoor coverage, which is vital for ensuring consistent network performance in indoor scenarios.	Complexity ,Environmantal variability , Frequency dependancy and accurate modeling often requires detaied data.	Status: Ongoing development and calibration of radio transmission into buildings.	Introducing field strength-based radio transmission modeling for building penetration. These novel ways will be used in the proposed project..	Expand model generalization through more diverse building measurements and refine calibration techniques.	Approach to calculate radio transmission into buildings based on field strength coverage at outer walls, with the development and calibration of various propagation models.	[5]SofhYare Tool Winprop for the Planning of Mobile Communication Networks: http ://wi nprop . ihf.uni - Stuttgart. de
Further Validation of an Electromagnetic Macro Model for Analysis of Propagation Path Loss in Cellular Networks Using Measured Driving-Test Data	M. N. Abdallah, IN. Dyab, T. K. Sarkar, M. V. s. N. Prasad, c. s. Misra, A. Lamparez, M. Salazar-Palmctl, and S. IN. Ting	Cost-Efficiency: Using electromagnetic macro modeling eliminates the need for expensive and time-consuming driving tests, reducing operational costs.	Initial Setup Complexity: Developing the electromagnetic macro model may require a significant effort in terms of data collection	Invention for macro modeling-based simulation in wireless communication	Using Macro modeling-based simulation in wireless communication technology These novel ways will be used in the proposed project	Develop new methods for validating the accuracy of macro modeling-based simulations.	develop a propagation model for wireless communications channels using macro modeling-based simulation	[6]M. Lazarus, "The Great Spectrum Famine," IEEE Spectrum, 47, 1 0, October 20 I 0, pp. 26-3 1.
Study And Implementation Of Drive Test For Development Of GSM Network	Richa Budhiraja1, Jitendra Singh Jadon2	Quality of Service (QoS) Evaluation: Drive tests provide a comprehensive evaluation of the network's QoS, including call quality, data performance, and network responsiveness	Limited Real-Time Data: Drive tests provide a snapshot of network performance at specific times and locations	related to drive testing in network planning	provided text lies in emphasizing the importance of drive testing as a key tool for assessing and maintaining the quality and coverage of mobile radio networks, particularly within the competitive and rapidly expanding	Advanced Data Analytics ,Develop real-time monitoring solutions to complement drive testing for immediate issue identification and resolution.	network planning and quality assessment approach that leverages drive testing as a fundamental tool in the GSM and mobile radio industry,	[R. Gahleitner: „Radio Wave Propagation in and into Urban Buildings", PhD-Thesis, TU Vienna, May 1994.7]

erformance of a Three-Dimensional Propagation Model in Urban Environments	M. Feistel, A. Baier Mannesmann Mobilfunk GmbH, Am Seestern 1,50453 Dusseldorf, Germany	Three-Dimensional Prediction Model: The use of a three-dimensional propagation model provides a more accurate representation of field strength prediction at 900 MHz in urban areas, accounting for both vertical and horizontal components.	Complexity & Database Dependency: The model's accuracy is influenced by the quality and accuracy of the database used, which can be a limitation in areas with incomplete or outdated data.	Ongoing research and developme nt.	Novelty lies in street patterns, building structures, and vector data.This novel ways will not be used in the proposed project	Enhance Urban-Micro-Model considering street patterns and building layouts.Investi gate improved data formats for more accurate predictions.	This research highlights the significance of street patterns and antenna heights in urban wave propagation	[3]R.J.Luebbers, "Finite conductivity uniform GTDversus knife edge diffraction", IEEE Trans. on An-tennas and Propgation, v01.32, no. 1, pp.70-76,1984

14.0)APPENDICES

14.1)Codes and libraries

14.1.1)Vivo Outside of the building

```
% Sample data (replace with your actual data)

rsrp_dBm = [-71 -72 -70];

rsrq = [-12 -13 -11];

snr = [3.5 4 2.8];


% Plot RSRP

figure(1);

plot(rsrp_dBm);
```

```
xlabel('Sample Number');
ylabel('RSRP (dBm)');
title('RSRP Variation');

% Plot RSRQ
figure(2);
plot(rsrq);
xlabel('Sample Number');
ylabel('RSRQ');
title('RSRQ Variation');

% Plot SNR
figure(3);
plot(snr);
xlabel('Sample Number');
ylabel('SNR');
title('SNR Variation');
```

14.1.2)Vivo Inside of the building

```
% Signal Parameters
```

```
RSRP = -92; % Received Signal Reference Power (dBm)

RSRQ = -14; % Reference Signal Received Quality

SNR = -2.0; % Signal-to-Noise Ratio (dB)

PCI = 324; % Physical Cell Identity

ARFCN = 1650; % Absolute Radio Frequency Channel Number

FDL = 1850.0; % Downlink Frequency (MHz)

FUL = 1755.0; % Uplink Frequency (MHz)

BW = 15; % Bandwidth (MHz)

TA = 1; % Timing Advance


% Analysis

disp('Signal Analysis:');

disp(['RSRP: ', num2str(RSRP), ' dBm']);

if RSRP >= -90 && RSRP <= -70

    disp('RSRP indicates good indoor reception.');

else

    disp('RSRP indicates poor reception.');

end

disp(['RSRQ: ', num2str(RSRQ)]);

if RSRQ > -19

    disp('RSRQ indicates excellent signal quality.');

elseif RSRQ > -14

    disp('RSRQ indicates good signal quality.');
```

```
elseif RSRQ > -9
    disp('RSRQ indicates average signal quality.');
else
    disp('RSRQ indicates poor signal quality.');
end
disp(['SNR: ', num2str(SNR), ' dB']);
if SNR > 5
    disp('SNR indicates good signal quality.');
else
    disp('SNR indicates poor signal quality.');
end
disp(['PCI: ', num2str(PCI)]);
disp(['ARFCN: ', num2str(ARFCN)]);
disp(['FDL: ', num2str(FDL), ' MHz']);
disp(['FUL: ', num2str(FUL), ' MHz']);
disp(['BW: ', num2str(BW), ' MHz']);
disp(['TA: ', num2str(TA)]);

% Plotting
figure;
subplot(2,1,1);
plot([RSRP, RSRQ, SNR], 'o-', 'LineWidth', 1.5);
title('Signal Parameters');
```

```
xlabel('Parameter');

ylabel('Value');

set(gca, 'XTick', 1:3, 'XTickLabel', {'RSRP', 'RSRQ', 'SNR'});

grid on;

subplot(2,1,2);

stem([PCI, ARFCN, FDL, FUL, BW, TA], 'LineWidth', 1.5);

title('Other Parameters');

xlabel('Parameter');

ylabel('Value');

set(gca, 'XTick', 1:6, 'XTickLabel', {'PCI', 'ARFCN', 'FDL', 'F
```

14.1.3)Samsung Outdoor result of the building

```
% Signal Parameters

RSRP = -92; % Received Signal Reference Power (dBm)

RSRQ = -14; % Reference Signal Received Quality

SNR = -2.0; % Signal-to-Noise Ratio (dB)

PCI = 324; % Physical Cell Identity

ARFCN = 1650; % Absolute Radio Frequency Channel Number

FDL = 1850.0; % Downlink Frequency (MHz)

FUL = 1755.0; % Uplink Frequency (MHz)

BW = 15; % Bandwidth (MHz)

TA = 1; % Timing Advance
```

```
% Analysis
disp('Signal Analysis:');
disp(['RSRP: ', num2str(RSRP), ' dBm']);
if RSRP >= -90 && RSRP <= -70
    disp('RSRP indicates good indoor reception.');
else
    disp('RSRP indicates poor reception.');
end
disp(['RSRQ: ', num2str(RSRQ)]);
if RSRQ > -19
    disp('RSRQ indicates excellent signal quality.');
elseif RSRQ > -14
    disp('RSRQ indicates good signal quality.');
elseif RSRQ > -9
    disp('RSRQ indicates average signal quality.');
else
    disp('RSRQ indicates poor signal quality.');
end
disp(['SNR: ', num2str(SNR), ' dB']);
if SNR > 5
    disp('SNR indicates good signal quality.');
else
```

```
    disp('SNR indicates poor signal quality.');
end
disp(['PCI: ', num2str(PCI)]);
disp(['ARFCN: ', num2str(ARFCN)]);
disp(['FDL: ', num2str(FDL), ' MHz']);
disp(['FUL: ', num2str(FUL), ' MHz']);
disp(['BW: ', num2str(BW), ' MHz']);
disp(['TA: ', num2str(TA)]);

% Plotting
figure;
subplot(2,1,1);
plot([RSRP, RSRQ, SNR], 'o-', 'LineWidth', 1.5);
title('Signal Parameters');
xlabel('Parameter');
ylabel('Value');
set(gca, 'XTick', 1:3, 'XTickLabel', {'RSRP', 'RSRQ', 'SNR'});
grid on;
subplot(2,1,2);
stem([PCI, ARFCN, FDL, FUL, BW, TA], 'LineWidth', 1.5);
title('Other Parameters');
xlabel('Parameter');
ylabel('Value');
```

```
set(gca, 'XTick', 1:6, 'XTickLabel', {'PCI', 'ARFCN', 'FDL',
'FUL', 'BW', 'TA'});
grid on;
```

14.1.4)Samsung Inside of the building result

```
% Signal Parameters
BAND = "EGSM"; % GSM network band
RSRP = -113; % Received Signal Reference Power (dBm)
SNR = 3.5; % Signal-to-Noise Ratio (dB)

% Analysis
disp('Signal Analysis:');
disp(['Band: ', BAND]);
disp(['RSRP: ', num2str(RSRP), ' dBm']);
if RSRP >= -70
    disp('RSRP indicates strong signal strength.');
elseif RSRP >= -90
    disp('RSRP indicates moderate signal strength.');
else
    disp('RSRP indicates weak signal strength.');
end
disp(['SNR: ', num2str(SNR), ' dB']);
if SNR > 5
```

```
    disp('SNR indicates good signal quality.');
else
    disp('SNR indicates poor signal quality.');
end

% Plotting
figure;
subplot(2,1,1);
plot(RSRP, 'o-', 'LineWidth', 1.5);
title('RSRP Analysis');
xlabel('Parameter');
ylabel('Value (dBm)');
set(gca, 'XTick', 1, 'XTickLabel', {'RSRP'});
grid on;
subplot(2,1,2);
plot(SNR, 'o-', 'LineWidth', 1.5);
title('SNR Analysis');
xlabel('Parameter');
ylabel('Value (dB)');
set(gca, 'XTick', 1, 'XTickLabel', {'SNR'});
grid on;
```

14.1.5)Galaxy J14 Inside and outside of the building

```
% Signal Parameters

BAND = "L1"; % LTE frequency band

RSRP = -92; % Received Signal Reference Power (dBm)

RSRQ = -13; % Reference Signal Received Quality (dB)

SNR = 0.0; % Signal-to-Noise Ratio (dB)

TECH = "LTE"; % Network technology

TAC = 50152; % Tracking Area Code

eNB = 20150; % eNodeB identifier

CELLID = 140; % Cell Identifier

PCI = 432; % Physical Cell Identity

ARFCN = 41292; % Absolute Radio Frequency Channel Number

FDL = 2660.2; % Downlink Frequency (MHz)

FUL = 39814.2; % Uplink Frequency (MHz)

BW = 15; % Bandwidth (MHz)


% Analysis

disp('Signal Analysis:');

disp(['Band: ', BAND]);

disp(['RSRP: ', num2str(RSRP), ' dBm']);

if RSRP >= -90

    disp('RSRP indicates good signal strength.');
```

```
elseif RSRP >= -100
    disp('RSRP indicates moderate signal strength.');
else
    disp('RSRP indicates weak signal strength.');
end
disp(['RSRQ: ', num2str(RSRQ), ' dB']);
if RSRQ > -19
    disp('RSRQ indicates excellent signal quality.');
elseif RSRQ > -14
    disp('RSRQ indicates good signal quality.');
elseif RSRQ > -9
    disp('RSRQ indicates average signal quality.');
else
    disp('RSRQ indicates poor signal quality.');
end
disp(['SNR: ', num2str(SNR), ' dB']);
if SNR > 5
    disp('SNR indicates good signal quality.');
else
    disp('SNR indicates poor signal quality.');
end
disp(['Technology: ', TECH]);
disp(['TAC: ', num2str(TAC)]);
```

```
disp(['eNB: ', num2str(eNB)]);

disp(['Cell ID: ', num2str(CELLID)]);

disp(['PCI: ', num2str(PCI)]);

disp(['ARFCN: ', num2str(ARFCN)]);

disp(['FDL: ', num2str(FDL), ' MHz']);

disp(['FUL: ', num2str(FUL), ' MHz']);

disp(['BW: ', num2str(BW), ' MHz']);
```

14.1.6)Result sheets of Inside the all floors using vivo phone 1 st floor

```
% Signal Parameters

BAND = "Not available"; % Band information not available

RSRP = -99; % Received Signal Reference Power (dBm)

RSRQ = -13; % Reference Signal Received Quality (dB)

SNR = 3; % Signal-to-Noise Ratio (dB)

TECH = "LTE"; % Network technology

CID = 140; % Cell Identifier

TAC = 50152; % Tracking Area Code

eNB = 20150; % eNodeB identifier

PCI = 432; % Physical Cell Identity
```

```
ARFCN = 41292; % Absolute Radio Frequency Channel Number

FDL = 2660.2; % Downlink Frequency (MHz)

FUL = 39814.2; % Uplink Frequency (MHz)

BW = 15; % Bandwidth (MHz)


% Analysis

disp('Signal Analysis:');

disp(['Band: ', BAND]);

disp(['RSRP: ', num2str(RSRP), ' dBm']);

if RSRP >= -90

    disp('RSRP indicates good signal strength.');

elseif RSRP >= -100

    disp('RSRP indicates moderate signal strength.');

else

    disp('RSRP indicates weak signal strength.');

end

disp(['RSRQ: ', num2str(RSRQ), ' dB']);

if RSRQ > -19

    disp('RSRQ indicates excellent signal quality.');

elseif RSRQ > -14

    disp('RSRQ indicates good signal quality.');

elseif RSRQ > -9

    disp('RSRQ indicates average signal quality.');
```

```
else
    disp('RSRQ indicates poor signal quality.');
end
disp(['SNR: ', num2str(SNR), ' dB']);
if SNR > 5
    disp('SNR indicates good signal quality.');
else
    disp('SNR indicates poor signal quality.');
end
disp(['Technology: ', TECH]);
disp(['CID: ', num2str(CID)]);
disp(['TAC: ', num2str(TAC)]);
disp(['eNB: ', num2str(eNB)]);
disp(['PCI: ', num2str(PCI)]);
disp(['ARFCN: ', num2str(ARFCN)]);
disp(['FDL: ', num2str(FDL), ' MHz']);
disp(['FUL: ', num2str(FUL), ' MHz']);
disp(['BW: ', num2str(BW), ' MHz']);
```

14.1.7)Result sheets of Inside the all floors using vivo phone 2nd floor

```
% Signal Parameters
RSRP = -105; % Received Signal Reference Power (dBm)
```

```
RSRQ = -14; % Reference Signal Received Quality (dB)

SNR = 11.5; % Signal-to-Noise Ratio (dB)

CELLID = 7; % Cell Identifier

PCE = 267; % Primary Cell Energy


% Analysis

disp('Signal Analysis:');

disp(['RSRP: ', num2str(RSRP), ' dBm']);

if RSRP >= -90 && RSRP <= -70

    disp('RSRP indicates good indoor reception.');

elseif RSRP > -70

    disp('RSRP indicates excellent indoor reception.');

else

    disp('RSRP indicates poor indoor reception. Occasional call
drops or slow data speeds may occur.');

end

disp(['RSRQ: ', num2str(RSRQ), ' dB']);

if RSRQ > -19

    disp('RSRQ indicates excellent signal quality.');

elseif RSRQ > -14

    disp('RSRQ indicates good signal quality.');

elseif RSRQ > -9

    disp('RSRQ indicates average signal quality.');
```

```
else
    disp('RSRQ indicates poor signal quality.');
end
disp(['SNR: ', num2str(SNR), ' dB']);
if SNR > 5
    disp('SNR indicates good signal quality.');
else
    disp('SNR indicates poor signal quality.');
end
disp(['Cell ID: ', num2str(CELLID)]);
disp(['PCE: ', num2str(PCE)]);

% Plotting
figure;
subplot(3,1,1);
plot(RSRP, 'o-', 'LineWidth', 1.5);
title('RSRP Analysis');
xlabel('Parameter');
ylabel('Value (dBm)');
set(gca, 'XTick', 1, 'XTickLabel', {'RSRP'});
grid on;
subplot(3,1,2);
plot(RSRQ, 'o-', 'LineWidth', 1.5);
```

```
title('RSRQ Analysis');

xlabel('Parameter');

ylabel('Value (dB)');

set(gca, 'XTick', 1, 'XTickLabel', {'RSRQ'});

grid on;

subplot(3,1,3);

plot(SNR, 'o-', 'LineWidth', 1.5);

title('SNR Analysis');

xlabel('Parameter');

ylabel('Value (dB)');

set(gca, 'XTick', 1, 'XTickLabel', {'SNR'});

grid on;
```

14.1.8)Result sheets of Inside the all floors using vivo phone 3rd floor

```
BAND = "L3"; % LTE Band

RSRP = -90; % Received Signal Reference Power (dBm)

RSRQ = -13; % Reference Signal Received Quality (dB)

SNR = 7.0; % Signal-to-Noise Ratio (dB)

CELLID = 7; % Cell Identifier

PC = 267; % Primary Cell Energy

ARFCN = 1275; % Absolute Radio Frequency Channel Number
```

```
FOL = 1812.5; % Frequency Downlink (MHz)

FUL = 1717.5; % Frequency Uplink (MHz)

BW = 15; % Bandwidth (MHz)

TA = 12; % Timing Advance


% Analysis

disp('Signal Analysis:');

disp(['Band: ', BAND]);

disp(['RSRP: ', num2str(RSRP), ' dBm']);

if RSRP >= -70

    disp('RSRP indicates excellent indoor reception.');

elseif RSRP >= -90

    disp('RSRP indicates acceptable indoor reception.');

else

    disp('RSRP indicates poor indoor reception. Occasional call
drops or slow data speeds may occur.');

end

disp(['RSRQ: ', num2str(RSRQ), ' dB']);

if RSRQ > -19

    disp('RSRQ indicates excellent signal quality.');

elseif RSRQ > -14

    disp('RSRQ indicates good signal quality.');

elseif RSRQ > -9
```

```
    disp('RSRQ indicates average signal quality.');
else
    disp('RSRQ indicates poor signal quality.');
end
disp(['SNR: ', num2str(SNR), ' dB']);
if SNR > 5
    disp('SNR indicates good signal quality.');
else
    disp('SNR indicates poor signal quality.');
end
disp(['Cell ID: ', num2str(CELLID)]);
disp(['PC: ', num2str(PC)]);
disp(['ARFCN: ', num2str(ARFCN)]);
disp(['FOL: ', num2str(FOL), ' MHz']);
disp(['FUL: ', num2str(FUL), ' MHz']);
disp(['BW: ', num2str(BW), ' MHz']);
disp(['TA: ', num2str(TA)]);

% Plotting
figure;
subplot(3,1,1);
plot(RSRP, 'o-', 'LineWidth', 1.5);
title('RSRP Analysis');
```

```
xlabel('Parameter');
ylabel('Value (dBm)');
set(gca, 'XTick', 1, 'XTickLabel', {'RSRP'});
grid on;
subplot(3,1,2);
plot(RSRQ, 'o-', 'LineWidth', 1.5);
title('RSRQ Analysis');
xlabel('Parameter');
ylabel('Value (dB)');
set(gca, 'XTick', 1, 'XTickLabel', {'RSRQ'});
grid on;
subplot(3,1,3);
plot(SNR, 'o-', 'LineWidth', 1.5);
title('SNR Analysis');
xlabel('Parameter');
ylabel('Value (dB)');
set(gca, 'XTick', 1, 'XTickLabel', {'SNR'});
grid on;
```

14.1.9)Result sheets of Inside the all floors using vivo phone 4th floor

```
% Signal Parameters
BAND = "L3"; % LTE frequency band
```

```
RSRP = -92; % Received Signal Reference Power (dBm)

RSRQ = -12; % Reference Signal Received Quality (dB)

SNR = 3.5; % Signal-to-Noise Ratio (dB)

CELLID = 24; % Cell Identifier

PCI = 324; % Physical Cell Identity

ARFCN = 1650; % Absolute Radio Frequency Channel Number

FDL = 1850.0; % Downlink Frequency (MHz)

FUL = 1755.0; % Uplink Frequency (MHz)

BW = 15; % Bandwidth (MHz)

TA = 1; % Timing Advance


% Analysis

disp('Signal Analysis:');

disp(['Band: ', BAND]);

disp(['RSRP: ', num2str(RSRP), ' dBm']);

if RSRP >= -90

    disp('RSRP indicates good signal strength for basic voice
calls and data usage.');

elseif RSRP >= -100

    disp('RSRP indicates acceptable signal strength, but may lead
to occasional call drops or slow data speeds.');

else

    disp('RSRP indicates weak signal strength. Occasional call
drops or slow data speeds may occur.');
```

```
end
disp(['RSRQ: ', num2str(RSRQ), ' dB']);
if RSRQ > -19
    disp('RSRQ indicates excellent signal quality.');
elseif RSRQ > -14
    disp('RSRQ indicates good signal quality.');
elseif RSRQ > -9
    disp('RSRQ indicates average signal quality.');
else
    disp('RSRQ indicates poor signal quality.');
end
disp(['SNR: ', num2str(SNR), ' dB']);
if SNR > 5
    disp('SNR indicates acceptable signal quality.');
else
    disp('SNR indicates poor signal quality.');
end
disp(['Cell ID: ', num2str(CELLID)]);
disp(['PCI: ', num2str(PCI)]);
disp(['ARFCN: ', num2str(ARFCN)]);
disp(['FDL: ', num2str(FDL), ' MHz']);
disp(['FUL: ', num2str(FUL), ' MHz']);
disp(['BW: ', num2str(BW), ' MHz']);
```

```
disp(['TA: ', num2str(TA)]);


% Plotting
figure;
subplot(3,1,1);
plot(RSRP, 'o-', 'LineWidth', 1.5);
title('RSRP Analysis');
xlabel('Parameter');
ylabel('Value (dBm)');
set(gca, 'XTick', 1, 'XTickLabel', {'RSRP'});
grid on;
subplot(3,1,2);
plot(RSRQ, 'o-', 'LineWidth', 1.5);
title('RSRQ Analysis');
xlabel('Parameter');
ylabel('Value (dB)');
set(gca, 'XTick', 1, 'XTickLabel', {'RSRQ'});
grid on;
subplot(3,1,3);
plot(SNR, 'o-', 'LineWidth', 1.5);
title('SNR Analysis');
xlabel('Parameter');
ylabel('Value (dB)');
```

```
set(gca, 'XTick', 1, 'XTickLabel', {'SNR'});

grid on;
```

14.1.10)Result sheets of Inside the all floors using vivo phone 5th floor

```
% Signal Parameters

BAND = "L1"; % LTE frequency band

RSRP = -97; % Received Signal Reference Power (dBm)

RSRQ = -14; % Reference Signal Received Quality (dB)

SNR = -2.0; % Signal-to-Noise Ratio (dB)

TECH = "LTE"; % Network technology

TAC = 43612; % Tracking Area Code

eNB = 3007; % eNodeB identifier

CELLID = 24; % Cell Identifier

PCI = 267; % Physical Cell Identity

ARFCN = 50; % Absolute Radio Frequency Channel Number

FDL = 2115.0; % Downlink Frequency (MHz)

FUL = 1925.0; % Uplink Frequency (MHz)

BW = 10; % Bandwidth (MHz)


% Analysis

disp('Signal Analysis:');

disp(['Band: ', BAND]);
```

```
disp(['RSRP: ', num2str(RSRP), ' dBm']);
if RSRP >= -90
    disp('RSRP indicates acceptable signal strength, but may result in occasional call drops or slow data speeds.');
else
    disp('RSRP indicates poor signal strength, suggesting frequent call drops and slow data speeds.');
end
disp(['RSRQ: ', num2str(RSRQ), ' dB']);
if RSRQ > -19
    disp('RSRQ indicates excellent signal quality.');
elseif RSRQ > -14
    disp('RSRQ indicates good signal quality.');
elseif RSRQ > -9
    disp('RSRQ indicates average signal quality.');
else
    disp('RSRQ indicates poor signal quality.');
end
disp(['SNR: ', num2str(SNR), ' dB']);
if SNR > 5
    disp('SNR indicates acceptable signal quality.');
else
    disp('SNR indicates poor signal quality.');
```

```
end
disp(['Technology: ', TECH]);
disp(['TAC: ', num2str(TAC)]);
disp(['eNB: ', num2str(eNB)]);
disp(['Cell ID: ', num2str(CELLID)]);
disp(['PCI: ', num2str(PCI)]);
disp(['ARFCN: ', num2str(ARFCN)]);
disp(['FDL: ', num2str(FDL), ' MHz']);
disp(['FUL: ', num2str(FUL), ' MHz']);
disp(['BW: ', num2str(BW), ' MHz']);

% Plotting
figure;
subplot(3,1,1);
plot(RSRP, 'o-', 'LineWidth', 1.5);
title('RSRP Analysis');
xlabel('Parameter');
ylabel('Value (dBm)');
set(gca, 'XTick', 1, 'XTickLabel', {'RSRP'});
grid on;
subplot(3,1,2);
plot(RSRQ, 'o-', 'LineWidth', 1.5);
title('RSRQ Analysis');
```

```
xlabel('Parameter');

ylabel('Value (dB)');

set(gca, 'XTick', 1, 'XTickLabel', {'RSRQ'});

grid on;

subplot(3,1,3);

plot(SNR, 'o-', 'LineWidth', 1.5);

title('SNR Analysis');

xlabel('Parameter');

ylabel('Value (dB)');

set(gca, 'XTick', 1, 'XTickLabel', {'SNR'});

grid on;
```

14.1.11)Result sheets of Inside the all floors using vivo phone 6th floor

```
% Signal Parameters

BAND = "L3"; % LTE frequency band

RSRP = -74; % Received Signal Reference Power (dBm)

RSRQ = -13; % Reference Signal Received Quality (dB)

SNR = 10; % Signal-to-Noise Ratio (dB)

TA = 1; % Timing Advance
```

```
% Analysis
disp('Signal Analysis:');
disp(['Band: ', BAND]);
disp(['RSRP: ', num2str(RSRP), ' dBm']);
if RSRP >= -90
    disp('RSRP indicates excellent signal strength for indoor reception.');
elseif RSRP >= -70
    disp('RSRP indicates good signal strength for indoor reception.');
else
    disp('RSRP indicates poor signal strength for indoor reception. Occasional call drops or slow data speeds may occur.');
end
disp(['RSRQ: ', num2str(RSRQ), ' dB']);
if RSRQ > -19
    disp('RSRQ indicates excellent signal quality.');
elseif RSRQ > -14
    disp('RSRQ indicates good signal quality.');
elseif RSRQ > -9
    disp('RSRQ indicates average signal quality.');
else
    disp('RSRQ indicates poor signal quality.');
end
```

```
disp(['SNR: ', num2str(SNR), ' dB']);
if SNR > 5
    disp('SNR indicates good signal quality.');
else
    disp('SNR indicates poor signal quality.');
end
disp(['TA: ', num2str(TA)]);


% Plotting
figure;
subplot(3,1,1);
plot(RSRP, 'o-', 'LineWidth', 1.5);
title('RSRP Analysis');
xlabel('Parameter');
ylabel('Value (dBm)');
set(gca, 'XTick', 1, 'XTickLabel', {'RSRP'});
grid on;
subplot(3,1,2);
plot(RSRQ, 'o-', 'LineWidth', 1.5);
title('RSRQ Analysis');
xlabel('Parameter');
ylabel('Value (dB)');
set(gca, 'XTick', 1, 'XTickLabel', {'RSRQ'});
```

```
grid on;

subplot(3,1,3);

plot(SNR, 'o-', 'LineWidth', 1.5);

title('SNR Analysis');

xlabel('Parameter');

ylabel('Value (dB)');

set(gca, 'XTick', 1, 'XTickLabel', {'SNR'});

grid on;
```